

ELECTRONICS



ΠΑΝΕΠΙΣΤΗΜΙΟ  
ΔΥΤΙΚΗΣ ΑΤΤΙΚΗΣ  
UNIVERSITY OF WEST ATTICA

DEPARTMENT OF COMPUTER AND INFORMATION  
TECHNOLOGY AND COMPUTER ENGINEERING

PROJECT 3  
RECTIFICATION (SINGLE - DOUBLE)

**STUDENT DETAILS**

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**STUDENT SEMESTER :** 7 °

**STUDENT STATUS :** UNDERGRADUATE

**PROGRAMME OF STUDY :** UNIWA

**LABORATORY SECTION :** ELECT 17 FRIDAY 13:00 - 15:00

**LABORATORY TEACHERS :** TSALERA ELENA - DIAMANTOPOULOS MICHAEL

**DELIVERY DATE :** 16/12/2022

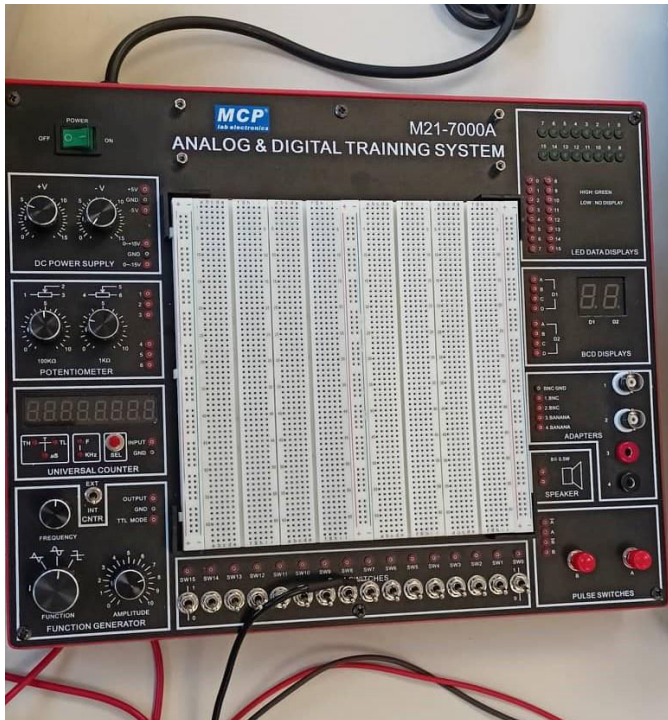
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**STUDENT PHOTO :**

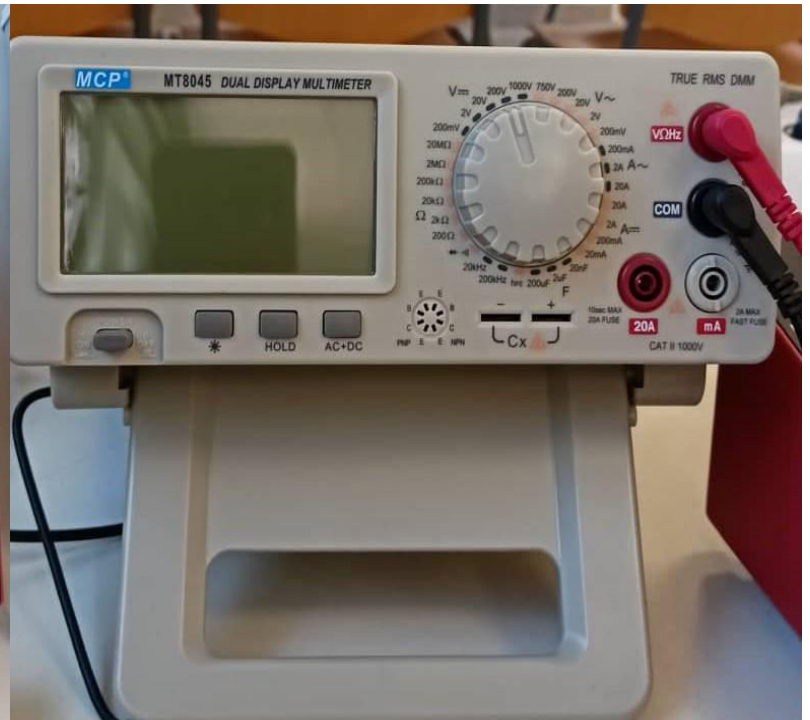


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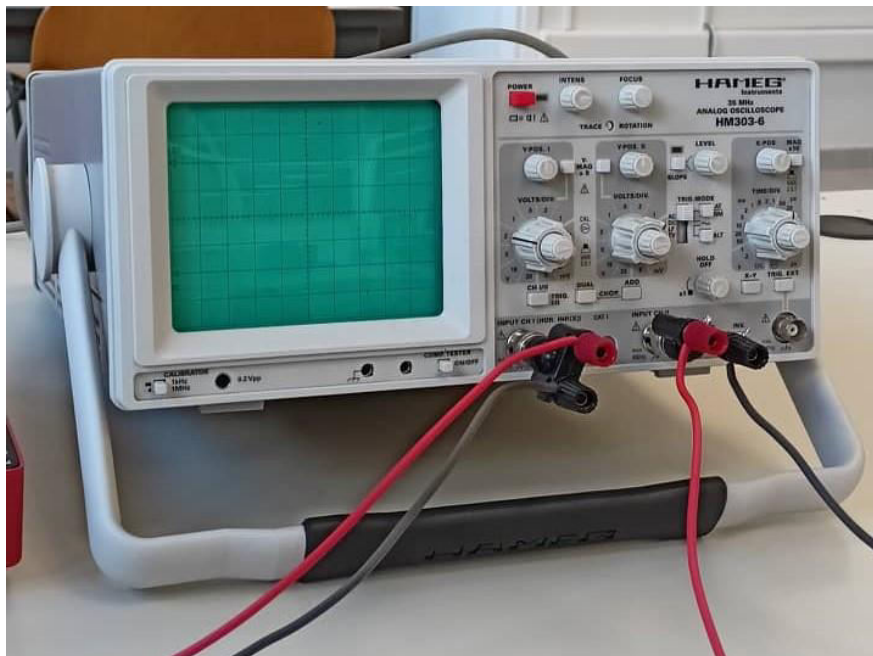
## PHOTOGRAPHS OF EQUIPMENT USED IN THE LABORATORY



*Breadboard M21-7000A*



*Digital Multimeter Bench MT8045*



*Oscilloscope HAMEG HM203-5*



*Cable*

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*Resistor of 4.7 KOhm*



*Silicon diode*

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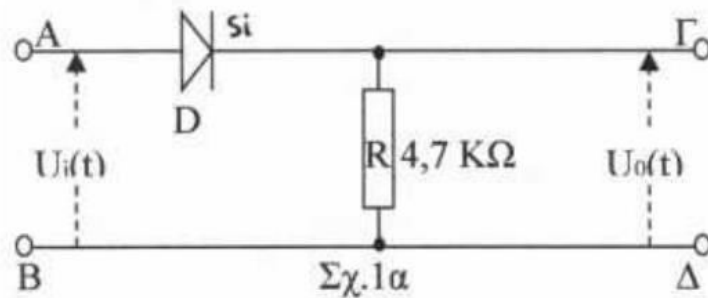
## 1. SIMPLE RECTIFICATION WITH RESISTANCE

### General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

### Theoretical Solving

Figure 1.1 shows the schematic illustration of the simple resistive lift experiment.



**Figure 1.1** Schematic illustration of resistance lifting

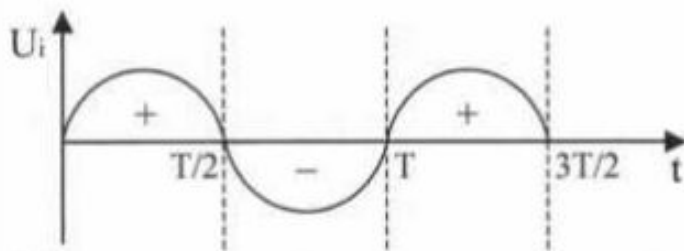
Table 1 lists the measurements regarding the voltage  $V_o$  in  $V_p$  calculated by the oscilloscope between signals C and D, the voltage  $V_\mu$  calculated by the voltmeter at these points and an average calculated value  $V_o / p$ . It is worth noting that an alternating voltage of  $8 V_{p-p}$  ( $4 V$ )<sub>p</sub> and a frequency of 1 KHz was applied to the circuit.

**Table 1**

| $V_{max}$ (Oscilloscope in V) <sub>p</sub> | $V_\mu$ (Voltmeter in V) | $V_\mu$ (V) = $V_{max} / \pi$ |
|--------------------------------------------|--------------------------|-------------------------------|
| 3.4                                        | 1.3                      | 1.08                          |

The average value of the 3<sup>rd</sup> column was calculated as follows :  $V_\mu = \frac{V_{max}}{\pi} \rightarrow V_\mu = \frac{3.4}{3.14} \rightarrow V_\mu = 1.08$  Volt and is less than the voltage calculated by the voltmeter in the 2<sup>nd</sup> column (1.3 Volt).

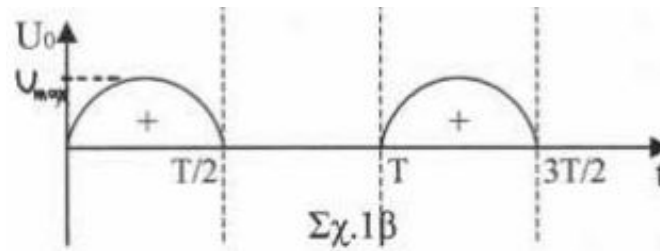
For the input signal  $V_i$  this waveform is formed as shown in Figure 1.2. As can be observed, it is a sine wave voltage with a period T supplied by an AC voltage source.



**Figure 1.2** Waveform of the input signal  $V_i$

For the output signal  $V_o$  this waveform is formed as shown in Figure 1.3. As observed, during the positive half-period of the input signal  $V_i$ , the diode turns on and current flows through it, as the voltage at the anode is higher than the voltage at the cathode. Hence the sine waveform formed by the output signal  $V_o$  during this half period. In contrast, during the negative half-period of the input signal  $V_i$ , the diode does not turn on and no current flows through it, as the voltage now at the cathode becomes greater than the voltage at the anode. Hence the absence of an output signal  $V_o$  during this half period.

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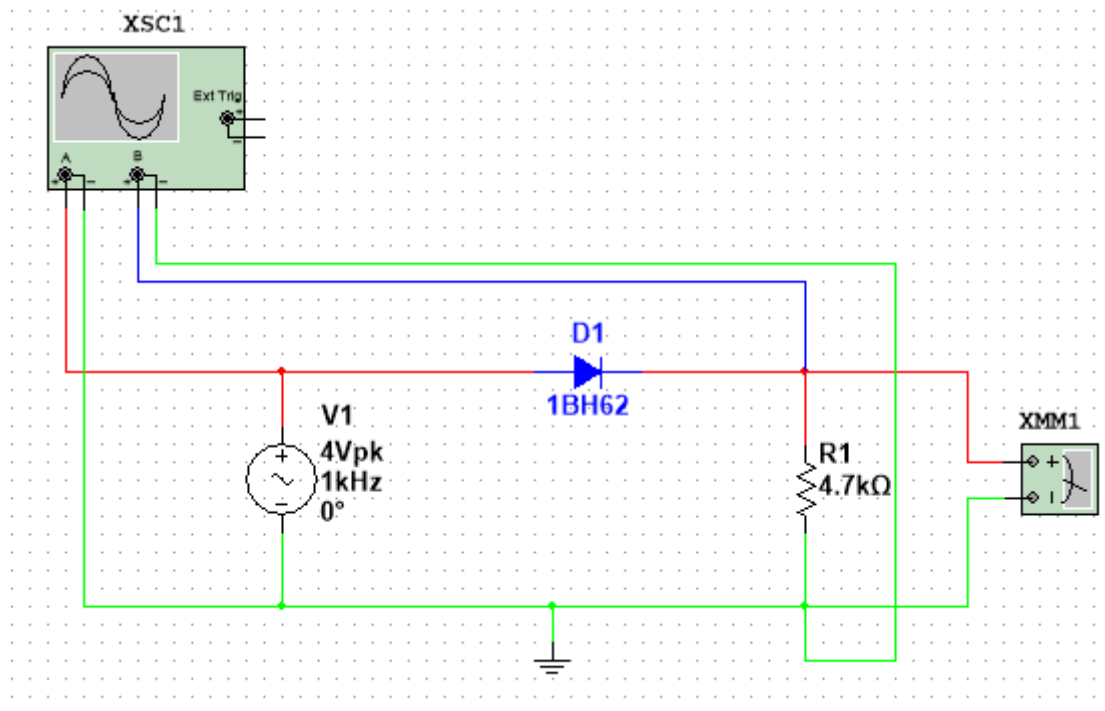


**Figure 1.3** Waveform of the output signal  $V_o$

In the output signal  $V_o$  there is also a voltage drop due to the 4.7 KOhm resistance of  $R$  and the silicon diode  $D$  from which the input signal  $V_i$  is passed to reach the ends of the resistor which is the output signal  $V$ .

## Simulative Resolution

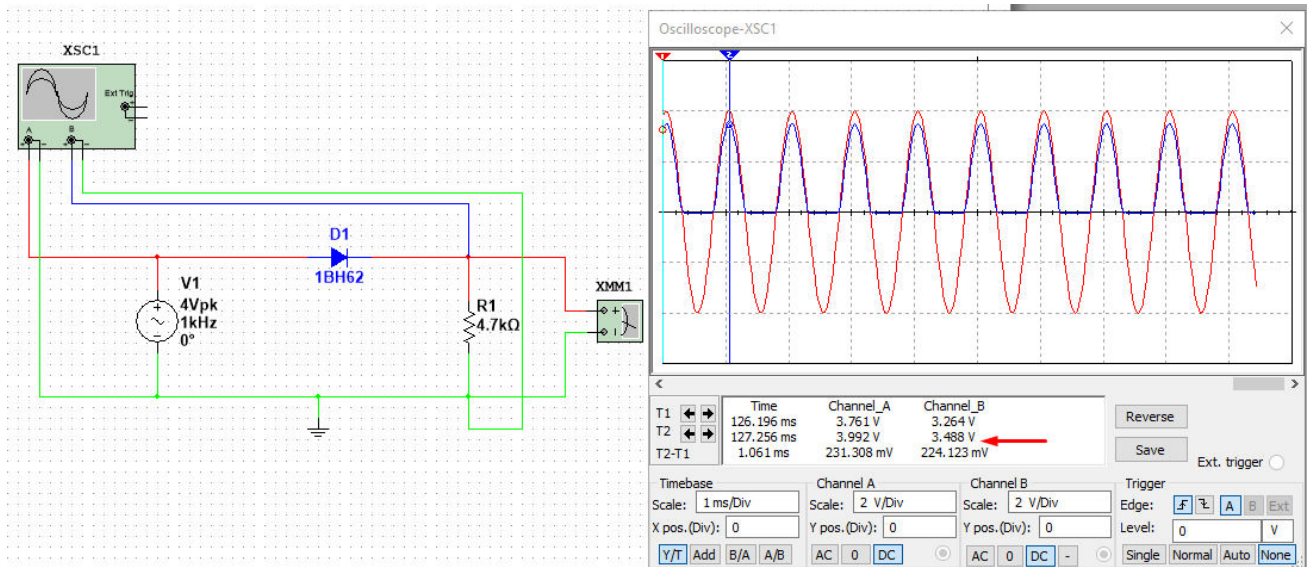
Figure 1.4 shows the simulation of the simple resistive rectification circuit in the simulation software "Multisim".



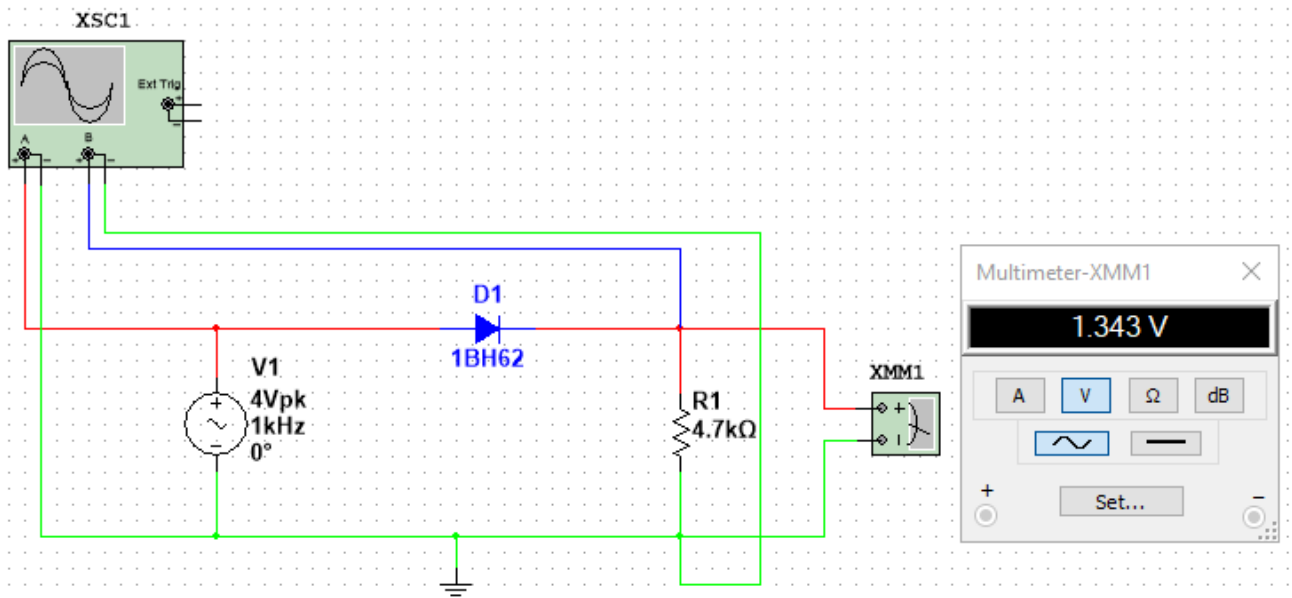
**Figure 1.4** Simulation of resistance recovery in "Multisim"

In Figures 1.5, 1.6 and 1.7, the recorded measurements of Table 1 are simulated in the simulation software.

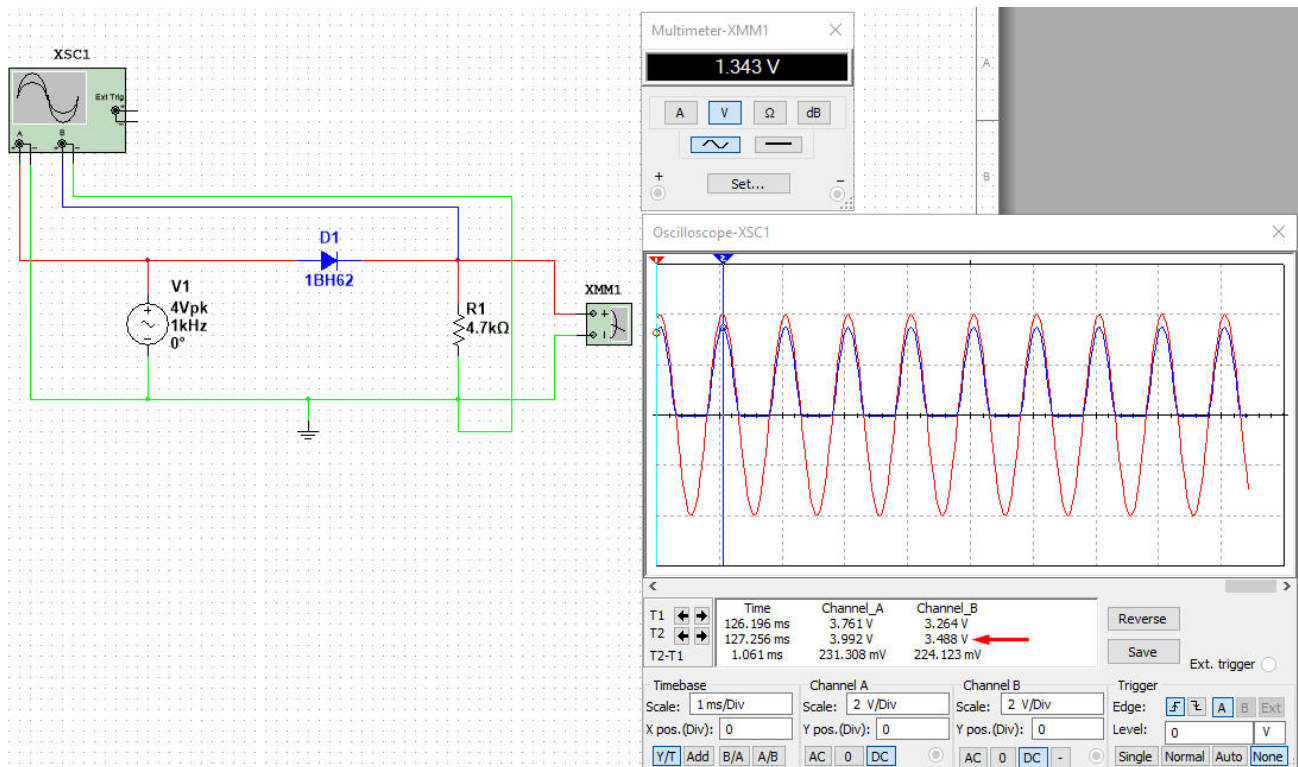
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**Figure 1.5** Calculation of  $V_{\max}$  of  $1^{\text{st}}$  column with the oscilloscope in "Multisim"



**Figure 1.6** Calculation of  $V_{\mu}$  of the  $2^{\text{nd}}$  column with the voltmeter in "Multisim"



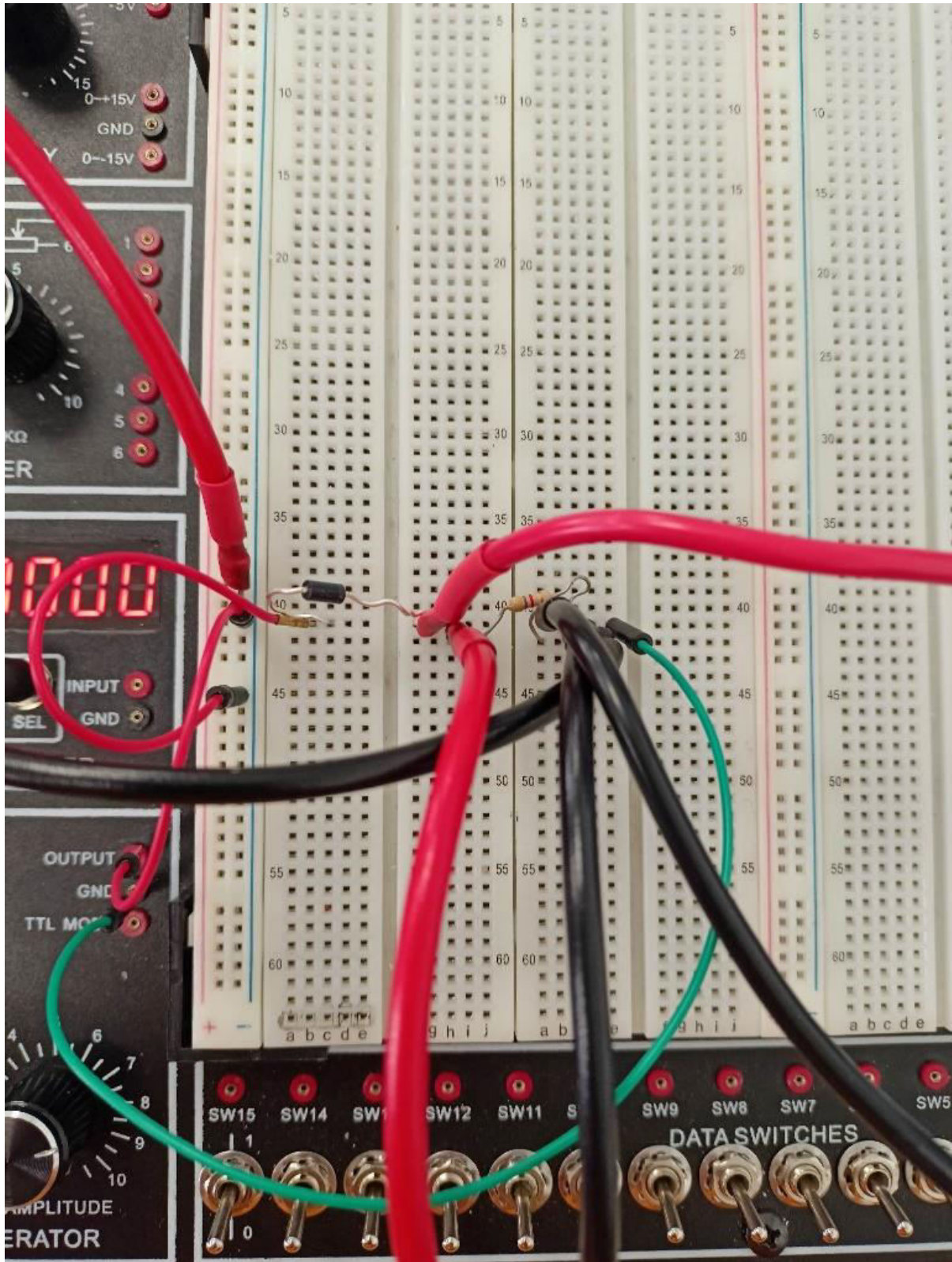
**Figure 1.7** Calculation of  $V_{\mu}$  of the 3<sup>th</sup> column with the voltmeter and oscilloscope in "Multisim"

## Experimental Solution

Figure 1.8 shows the circuit of the simple resistive rectification on the "breadboard", where it was carried out in the laboratory. The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 1.9).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 1.10).
3. Together with the power supply we connect the grounding cable to the breadboard (Figure 1.10).
4. We take channel A of the oscilloscope and connect the terminals to display the waveform on the screen. (Figure 1.10)
5. We set the switch to 2 Volts/Div. (Figure 1.11)
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 8 V<sub>p-p</sub> (Figure 1.11)
7. We align the waveform from the oscilloscope to be correctly positioned on the axes. (Figure 1.11)
8. We continue assembling the circuit by taking the silicon diode (Figure 1.12) and connecting it in series with the source with its cathode facing the output.
9. We take the 4.7 KOhm resistor (Figure 1.13) and connect it in series with the diode.
10. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the resistor to obtain the output signal  $V_{\max}$  of the 1<sup>th</sup> column of Table 1 (Figure 1.14).
11. We also connect the ends of the digital multimeter bench set to calculate alternating and continuous voltage in parallel with the ends of the resistor to obtain the voltage  $V_{\mu}$  of the 2<sup>th</sup> column in Table 1 (Figure 1.15).

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**Figure 1.8** Experimental illustration of the simple resistance lift on the "breadboard"

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Figure 1.9 Step 1

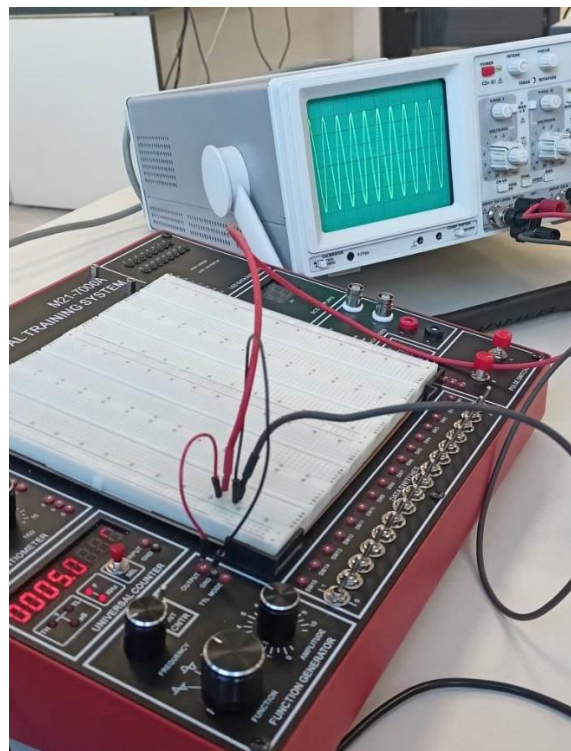


Figure 1.10 Step 2, 3, 4

# ELECTRONICS

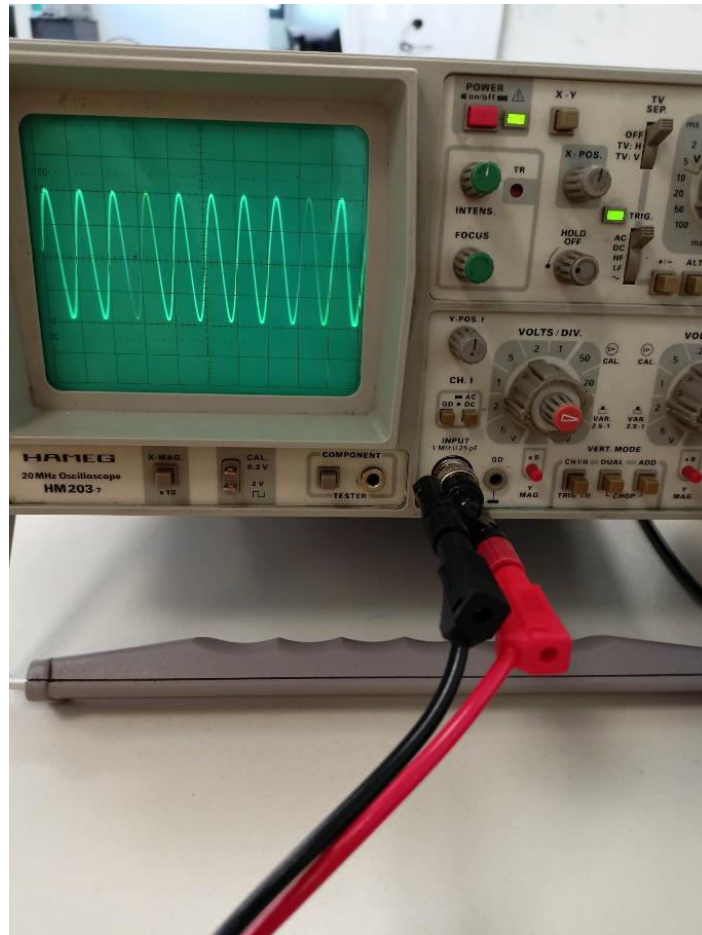


Figure 1.11 Step 5, 6, 7

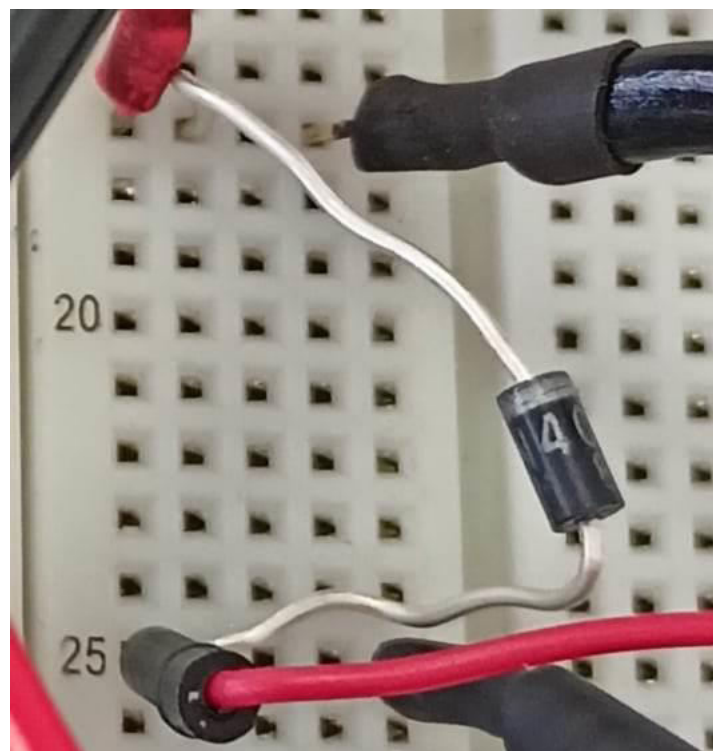
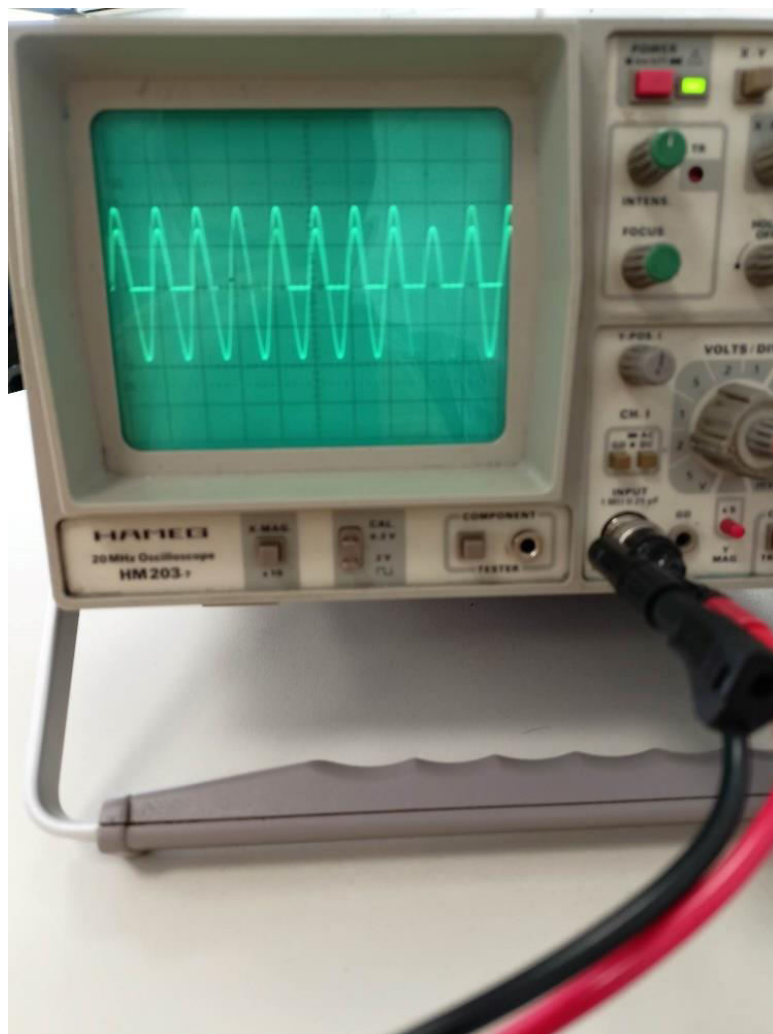


Figure 1.12 Step 8

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**Figure 1.13 Step 9**



**Figure 1.14 Step 10**

# ELECTRONICS

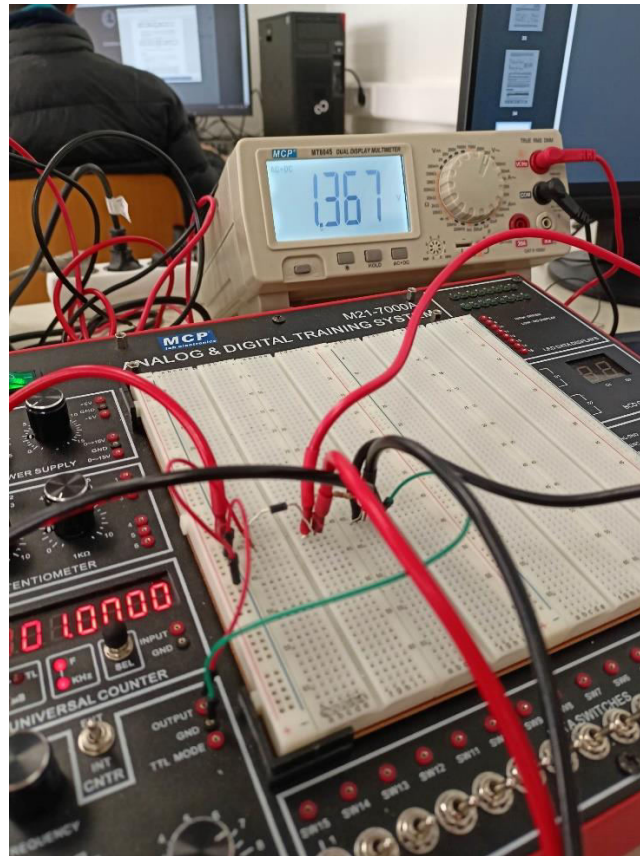


Figure 1.15 Step 11

## 2. SIMPLE RECTIFICATION WITH DIODE-SWITCH (PART A)

### General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

### Theoretical Solving

Figure 2.1 shows the schematic illustration of the simple rectification experiment with diode in the role of a switch, where the anode of the diode is directed towards nodes A and C.

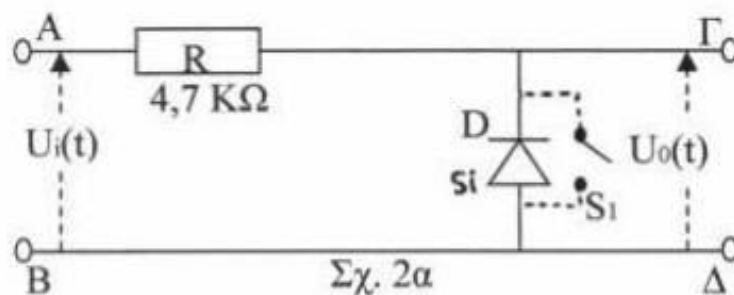


Figure 2.1 Schematic illustration of the rectification with a diode in the role of a switch with the anode facing nodes A and C

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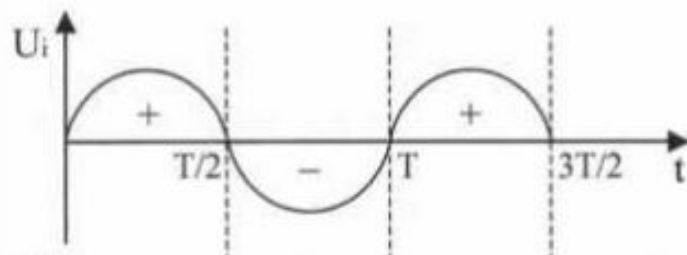
Table 2 lists the measurements regarding the voltage  $V_o$  in  $V_p$  calculated by the oscilloscope between signals C and D, the voltage  $V_\mu$  calculated by the voltmeter at these points and an average calculated value  $V_o / p$ . It is worth noting that an alternating voltage of  $8 V_{p-p}$  ( $4 V$ )<sub>p</sub> and a frequency of 1 KHz was applied to the circuit.

**Table 2**

| $V_{max}$ (Oscilloscope in V) <sub>p</sub> | $V_\mu$ (Voltmeter in V) | $V_\mu$ (V) = $V_{max} / \pi$ |
|--------------------------------------------|--------------------------|-------------------------------|
| 4                                          | 1.7                      | 1.3                           |

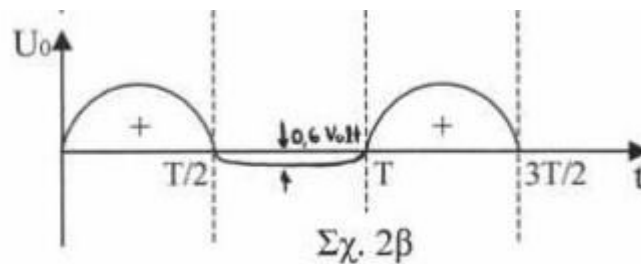
The average value of the 3<sup>th</sup> column was calculated as follows :  $V_\mu = \frac{V_{max}}{\pi} \rightarrow V_\mu = \frac{4}{3.14} \rightarrow V_\mu = 1.3$  Volt and is less than the voltage calculated by the voltmeter in the 2<sup>nd</sup> column (1.7 Volt).

For the input signal  $V_i$  this waveform is formed as shown in Figure 2.2. As observed, it is a sine wave voltage with a period T supplied by an AC voltage source.



**Figure 2.2** Waveform of the input signal  $V_i$

For the output signal  $V_o$  this waveform is formed as shown in Figure 2.3. As observed, during the positive half-period of the input signal  $V_i$ , the diode does not turn on and no current flows through it, as the voltage at the cathode is higher than the voltage at the anode. Therefore, the diode behaves like an open switch. Hence the sine waveform that forms the output signal  $V_o$  during this half period, which is the same as the waveform  $V_i$ . In contrast, during the negative half-period of the input signal  $V_i$ , the diode turns on and current flows through it, as the voltage now at the anode becomes higher than the voltage at the cathode. Therefore, the diode behaves like a closed switch. Points C and D short circuit and a voltage drop equal to the silicon diode blocking voltage (0.6 volts) occurs.

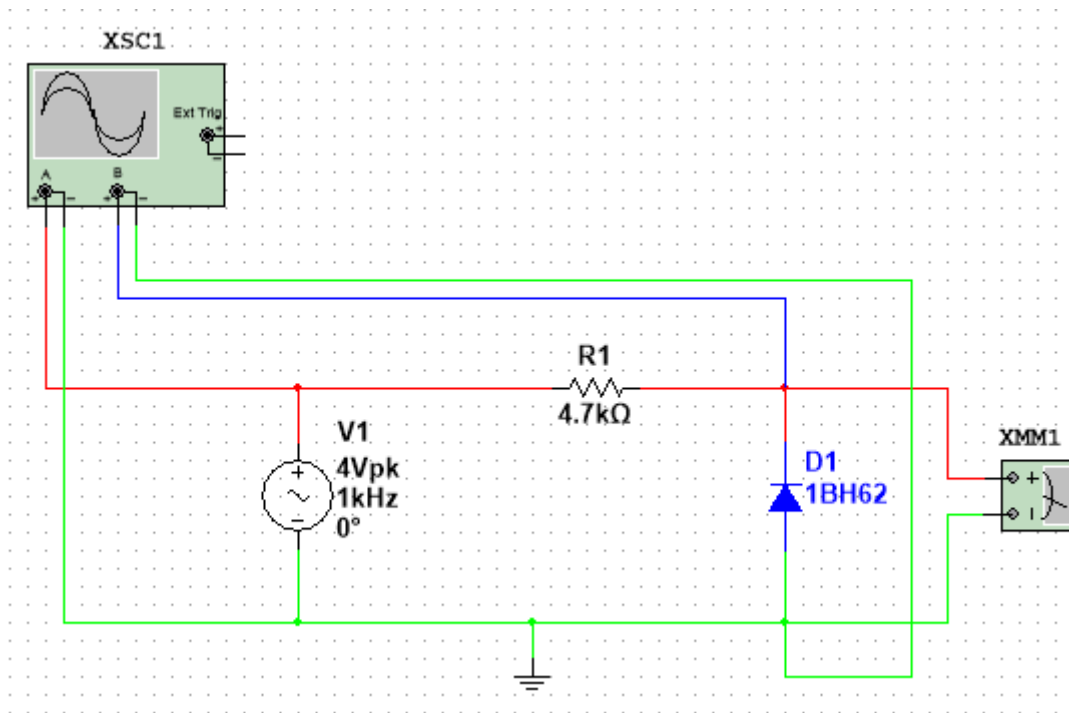


**Figure 2.3** Waveform of the input signal  $V_o$

## Simulative Resolution

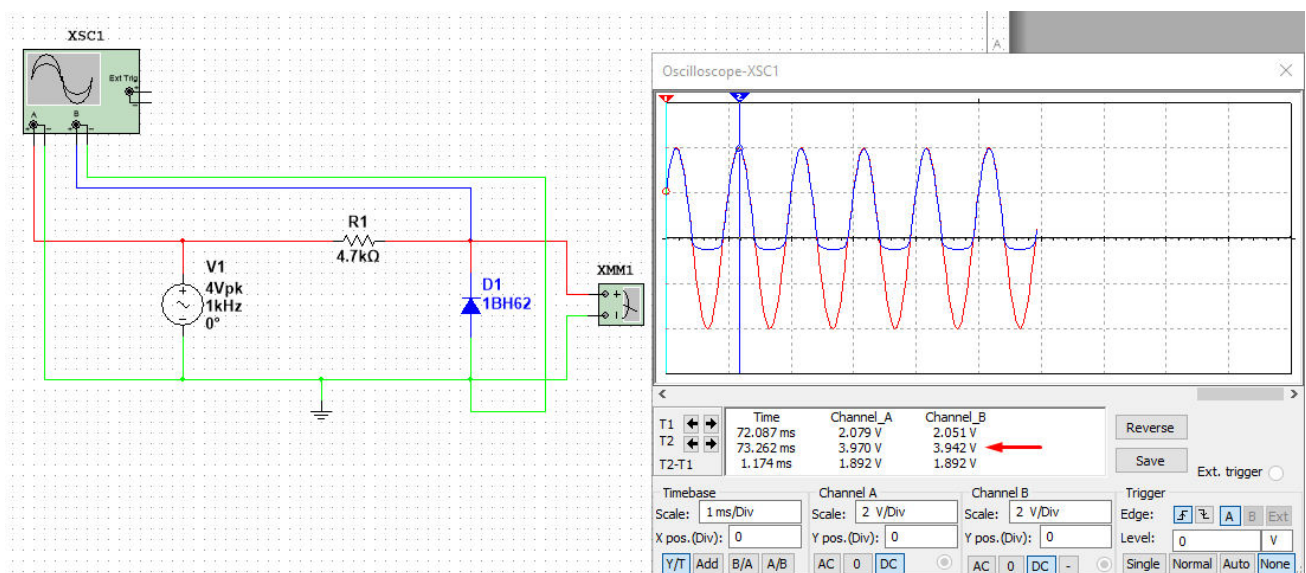
Figure 2.4 shows the simulation of the simple rectification circuit with a diode in the role of a switch with the anode pointing upwards, in the simulation software "Multisim".

# ELECTRONICS

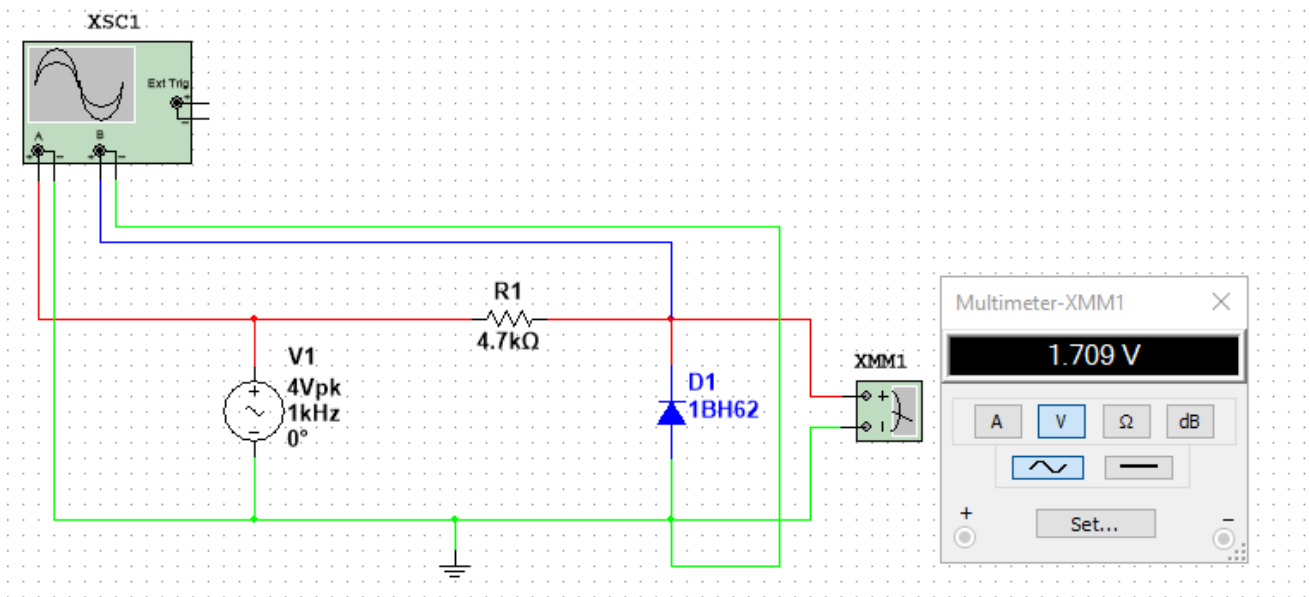


**Figure 2.4** Simulation of a simple lift with a diode in the role of a switch with the anode facing upwards in "Multisim"

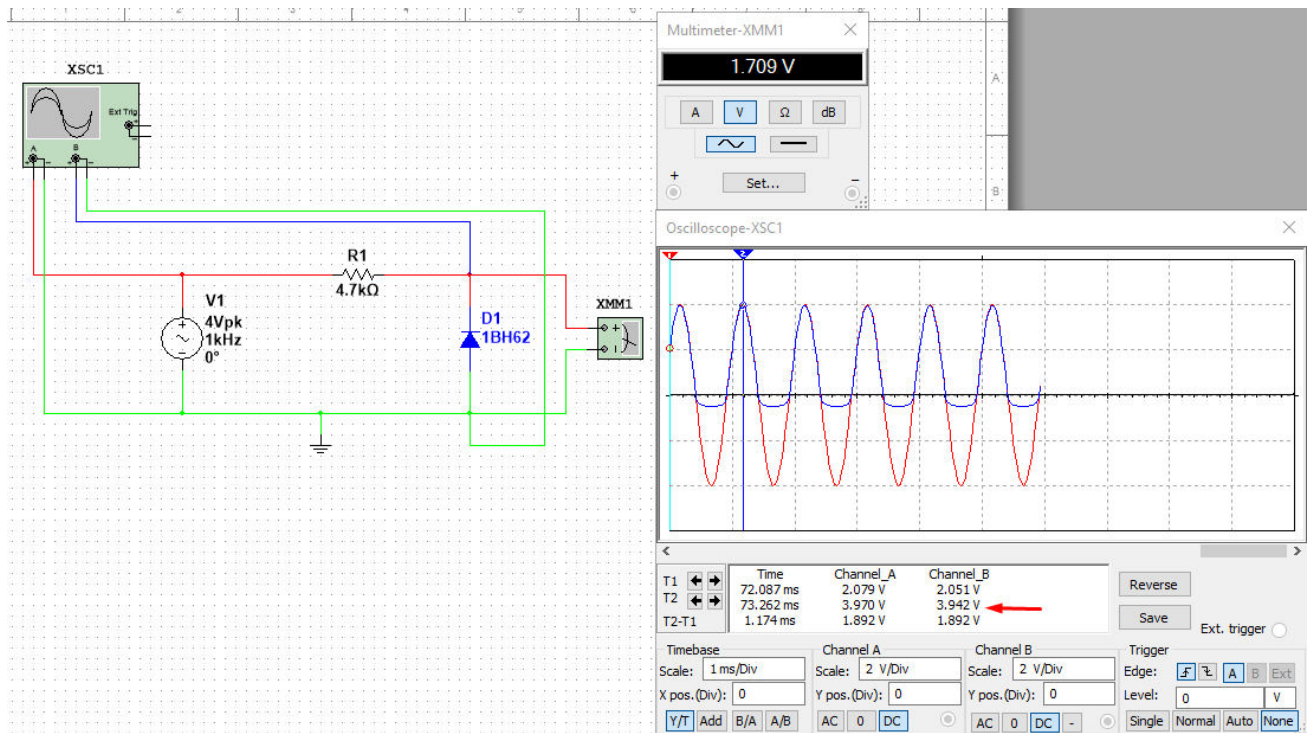
In Figures 2.5, 2.6 and 2.7, the recorded measurements of Table 2 are simulated in the simulation software.



**Figure 2.5** Calculation of  $V_{\max}$  of 1<sup>st</sup> column with the oscilloscope in "Multisim"



**Figure 2.6** Calculation of  $V_{\mu}$  of the 2<sup>nd</sup> column with the voltmeter in "Multisim"



**Figure 2.7** Calculation of  $V_{\mu}$  of the 3<sup>rd</sup> column with the voltmeter and oscilloscope in "Multisim"

## Experimental Solution

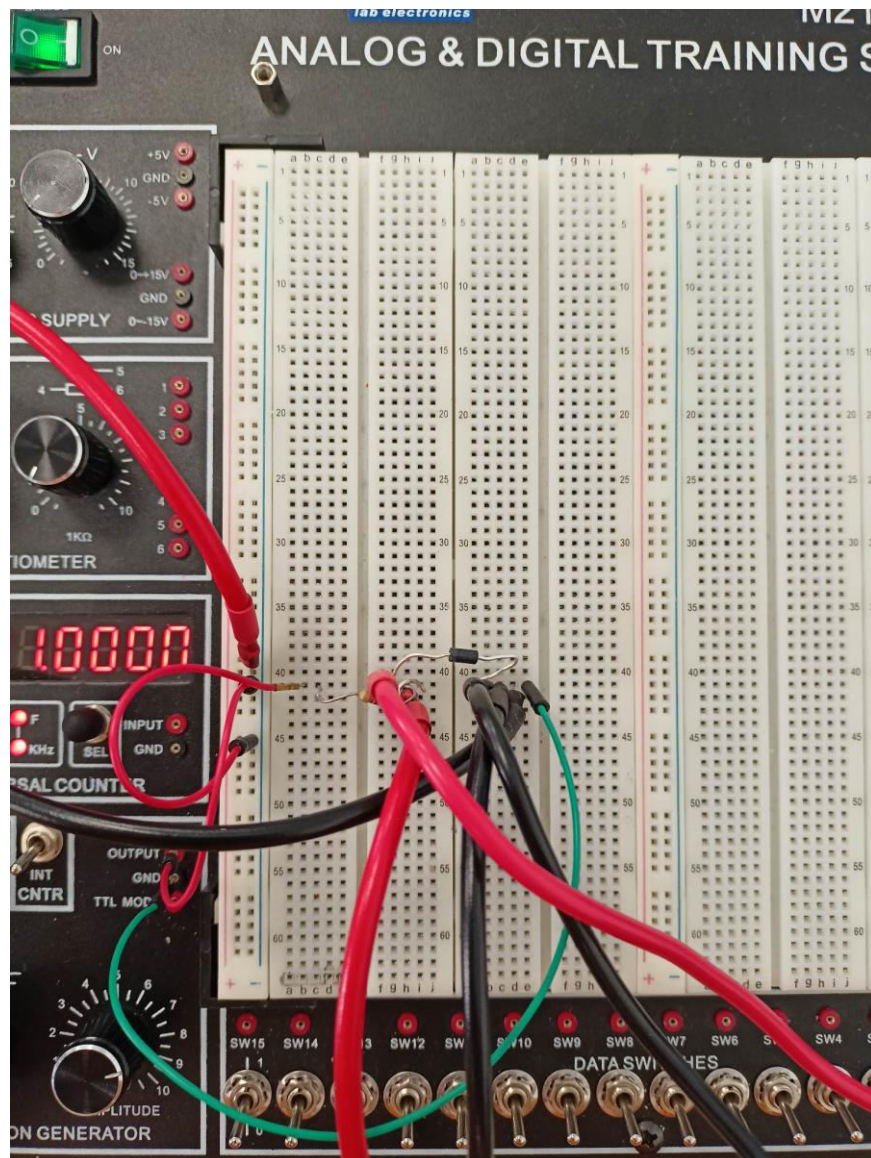
Figure 2.8 shows the circuit of the simple rectification with diode as a switch and the anode facing the source, on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 2.9).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 2.10).
3. Together with the power supply we connect the grounding cable to the breadboard (Figure 2.10).
4. We take channel A of the oscilloscope and connect the terminals to display the waveform on the screen. (Figure 2.10)

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5. We set the switch to 2 Volts/Div. (Figure 2.11)
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 8  $V_{p-p}$  (Figure 2.11)
7. We align the waveform from the oscilloscope to be correctly positioned on the axes. (Figure 2.11)
8. We continue assembling the circuit by taking the 4.7 KOhm resistor and connecting it in series with the source (Figure 2.12).
9. We take the silicon diode and connect it in series with the resistor, with the anode pointing towards the source (Figure 2.13)
10. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the diode to obtain the output signal  $V_{max}$  of the 1<sup>st</sup> column of Table 2 (Figure 2.14).
11. We also connect the ends of the digital multimeter bench set to calculate alternating and continuous voltage in parallel with the ends of the diode to obtain the voltage  $V_{\mu}$  of the 2<sup>nd</sup> column in Table 2 (Figure 2.15).



**Figure 2.8** Experimental illustration of simple rectification with a diode in the role of a switch, with the anode facing the source on the "breadboard"

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Figure 2.9 Step 1

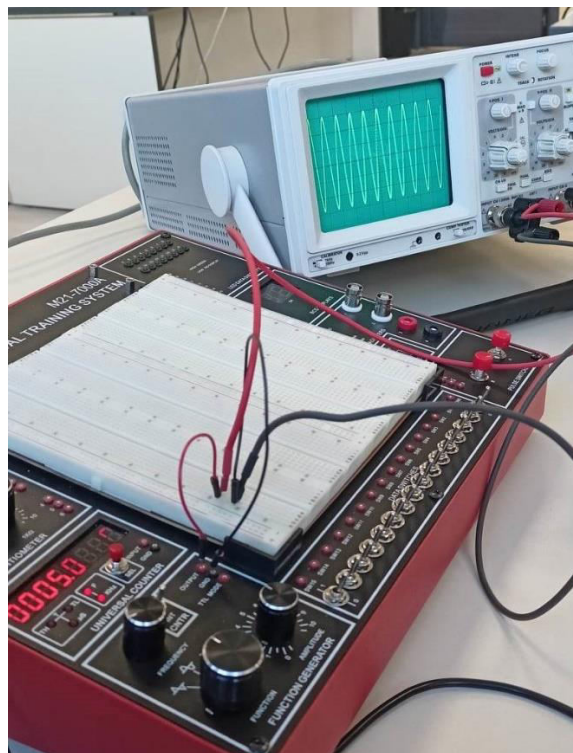


Figure 2.10 Step 2, 3, 4

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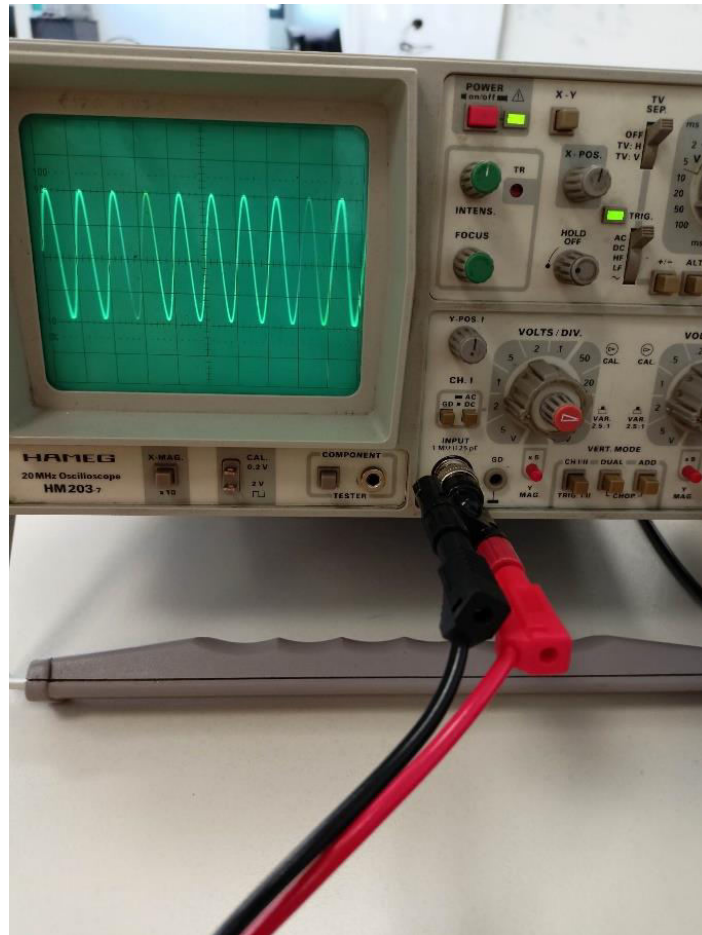


Figure 2.11 Step 5, 6, 7



Figure 2.12 Step 8

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Figure 2.13 Step 9

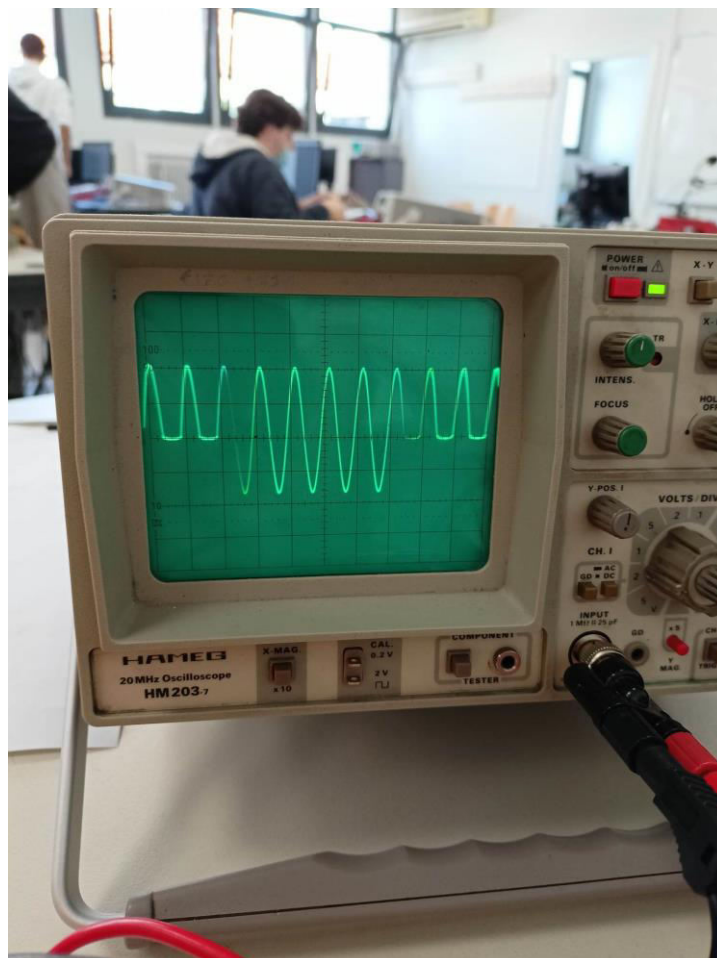


Figure 2.14 Step 10

# ELECTRONICS

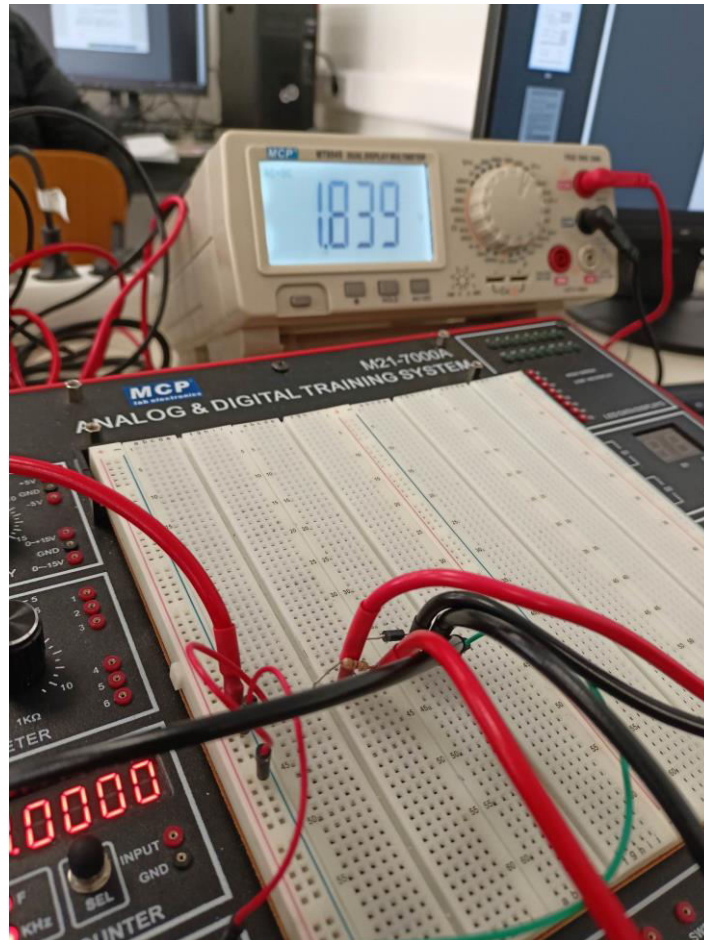


Figure 2.15 Step 11

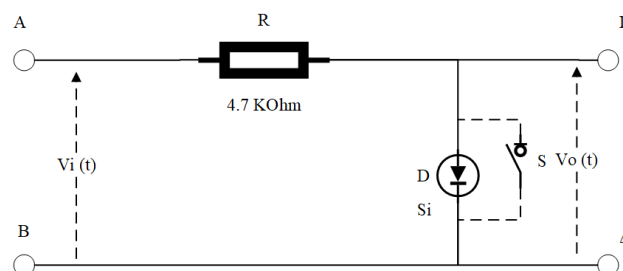
## 3. SIMPLE RECTIFICATION WITH DIODE-SWITCH (PART B)

### General

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

### Theoretical Solving

Figure 3.1 shows the schematic of the simple rectification experiment with diode in the switch role, where the anode of the diode is directed towards nodes B and D.



**Figure 3.1** Schematic illustration of the rectification with a diode in the role of a switch with the anode facing nodes B and D

# ELECTRONICS

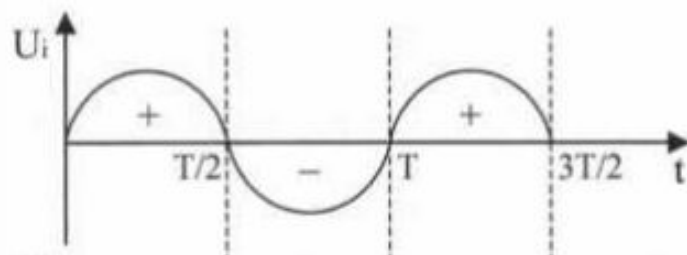
Table 3 lists the measurements regarding the voltage  $V_o$  in  $V_p$  calculated by the oscilloscope between signals C and D, the voltage  $V_\mu$  calculated by the voltmeter at these points and an average calculated value  $V_o / p$ . It is worth noting that an alternating voltage of  $8 V_{p-p}$  ( $4 V$ )<sub>p</sub> and a frequency of 1 KHz was applied to the circuit.

**Table 3**

| $V_{max}$ (Oscilloscope in V) <sub>p</sub> | $V_\mu$ (Voltmeter in V) | $V_\mu$ (V) = $V_{max} / \pi$ |
|--------------------------------------------|--------------------------|-------------------------------|
| 4                                          | 1.7                      | 1.3                           |

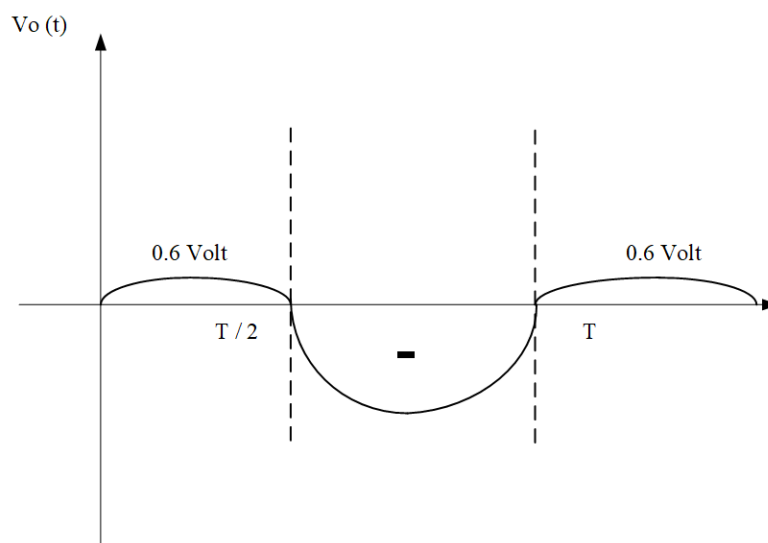
The average value of the 3<sup>th</sup> column was calculated as follows :  $V_\mu = \frac{V_{max}}{\pi} \rightarrow V_\mu = \frac{4}{3.14} \rightarrow V_\mu = 1.3 \text{ Volt}$  and is less than the voltage calculated by the voltmeter in the 2<sup>nd</sup> column (1.7 Volt).

For the input signal  $V_i$  this waveform is formed as shown in Figure 3.2. As observed, it is a sinusoidal voltage with a period T supplied by an AC voltage source.



**Figure 3.2** Waveform of the input signal  $V_i$

For the output signal  $V_o$ , this waveform is formed as shown in Figure 3.3. As observed, during the positive half-period of the input signal  $V_i$ , the diode turns on and current flows through it, as the voltage at the cathode is lower than the voltage at the anode. Therefore, the diode behaves like a grounded switch. Points C and D short-circuit and a voltage drop equal to the barrier voltage of the silicon diode (0.6 Volt) occurs. Hence the sine waveform that forms the output signal  $V_o$  during this half-period, which is the same as the  $V_i$  waveform. In contrast, during the negative half-period of the input signal  $V_i$ , the diode does not turn on and no current flows through it, as the voltage now at the anode becomes lower than the voltage at the cathode. Therefore, the diode behaves like an open switch. Hence the sine waveform that forms the output signal  $V_o$  during this half period, which is the same as the waveform  $V_i$ .

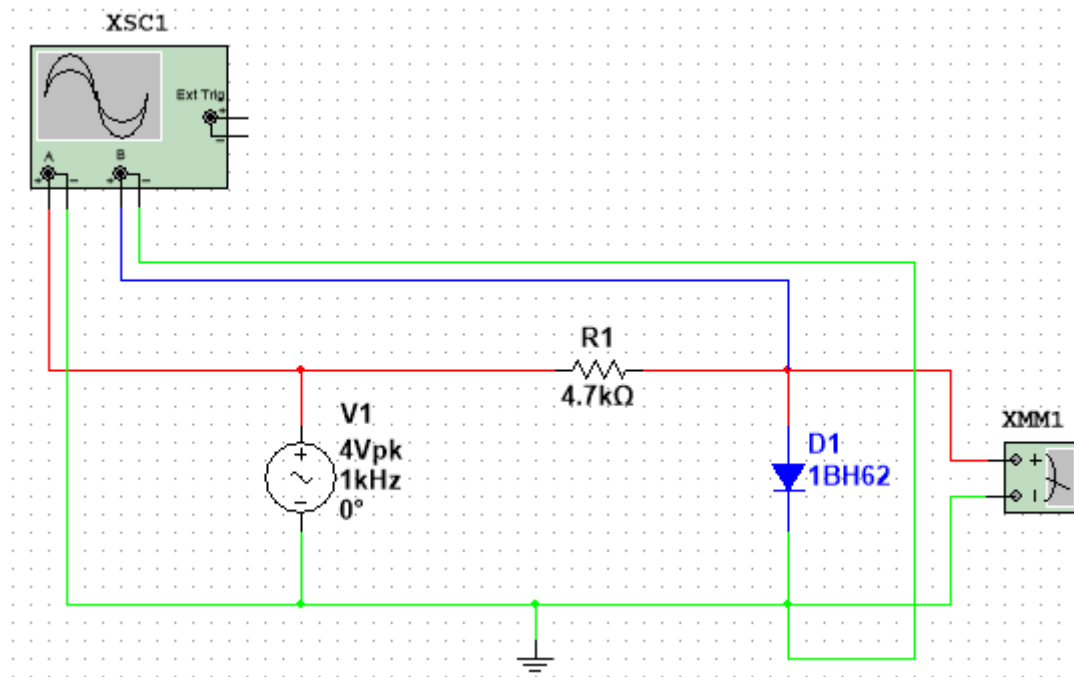


**Figure 3.3** Waveform of the input signal  $V_o$

## Simulative Resolution

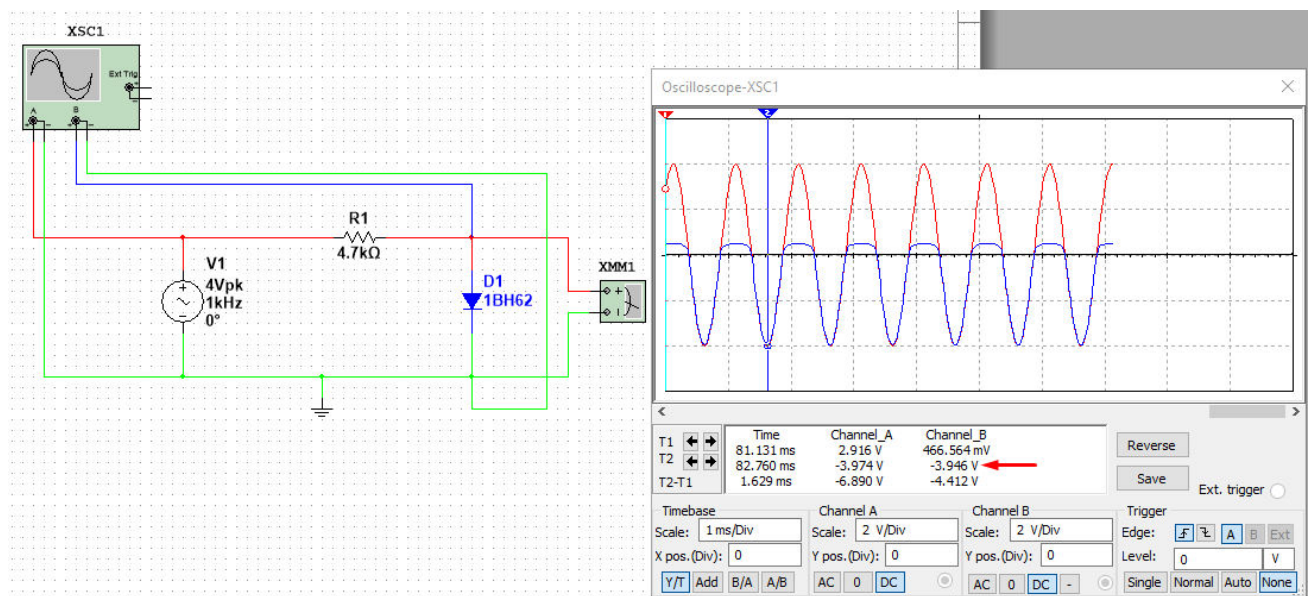
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Figure 3.4 shows the simulated illustration of the simple rectification circuit with a diode in the role of a switch with the anode pointing downwards, in the simulation software "Multisim".



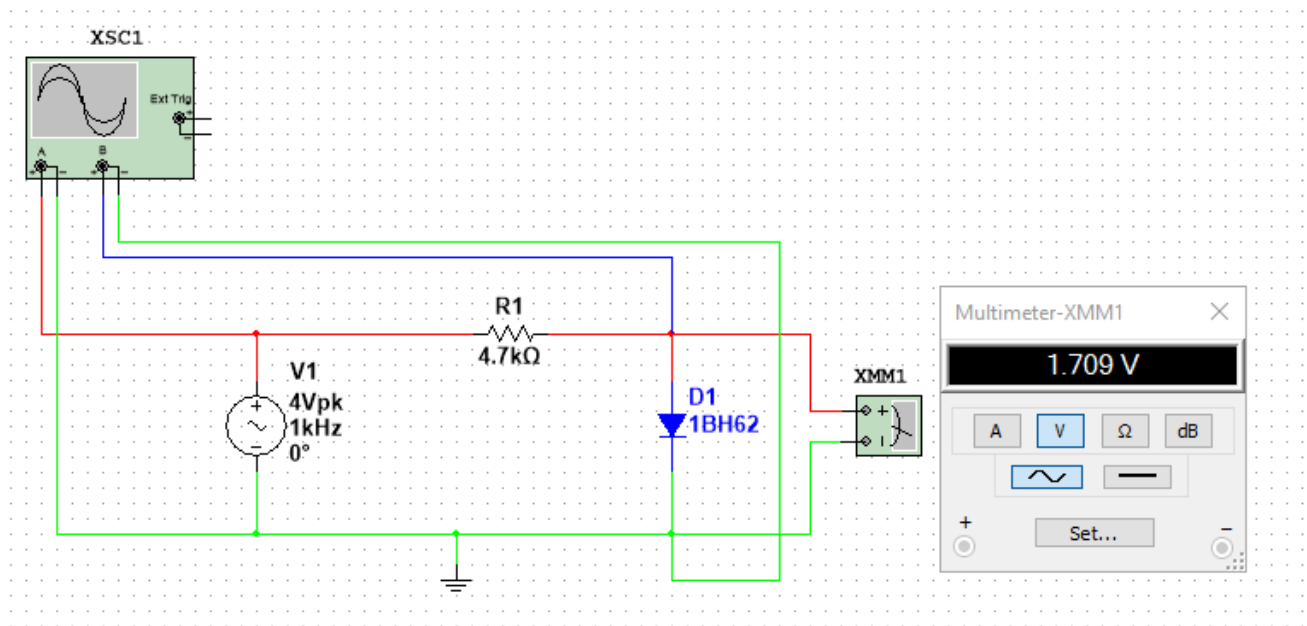
**Figure 3.4** Simulation of a simple lift with a diode in the role of a switch with the anode facing downwards in "Multisim"

In Figures 3.5, 3.6 and 3.7, the recorded measurements of Table 3 are simulated in the simulation software.

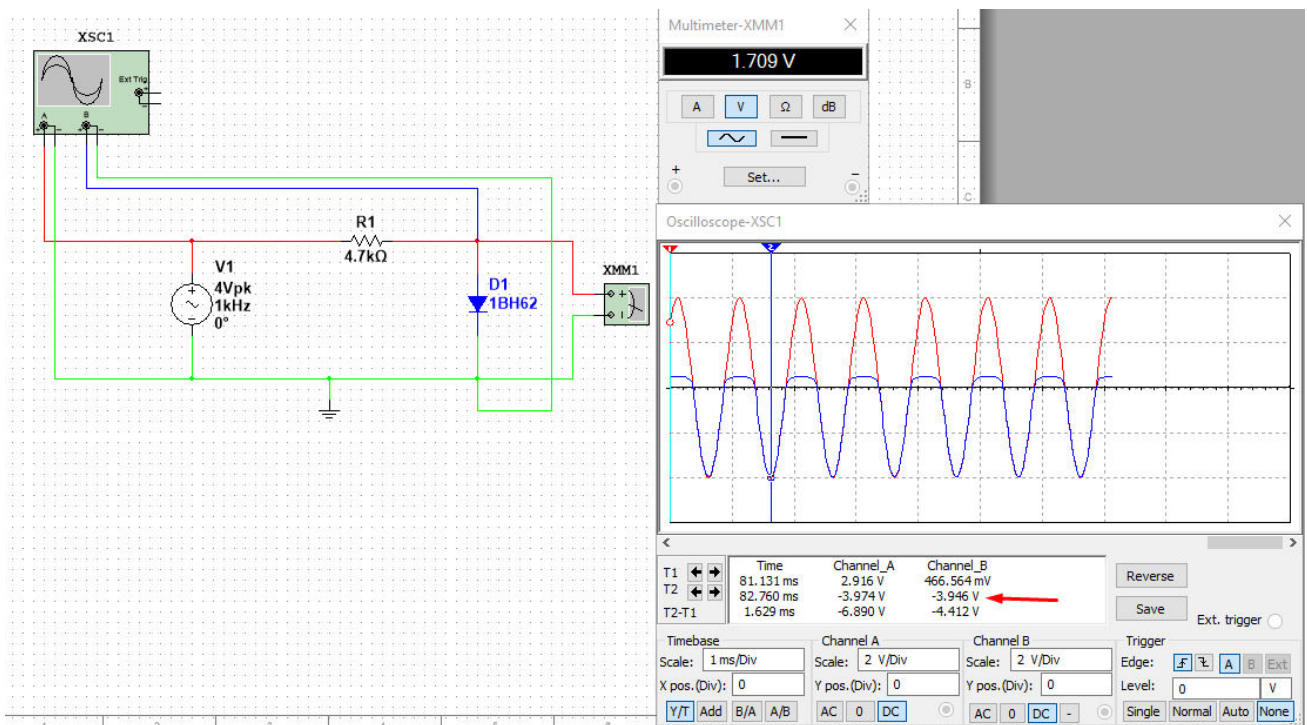


**Figure 3.5** Calculation of  $V_{\max}$  of 1<sup>th</sup> column with the oscilloscope in "Multisim"

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**Figure 3.6** Calculation of  $V_{\mu}$  of the 2<sup>ns</sup> column with the voltmeter in "Multisim"



**Figure 3.7** Calculation of  $V_{\mu}$  of the 3<sup>ns</sup> column with the voltmeter and oscilloscope in "Multisim"

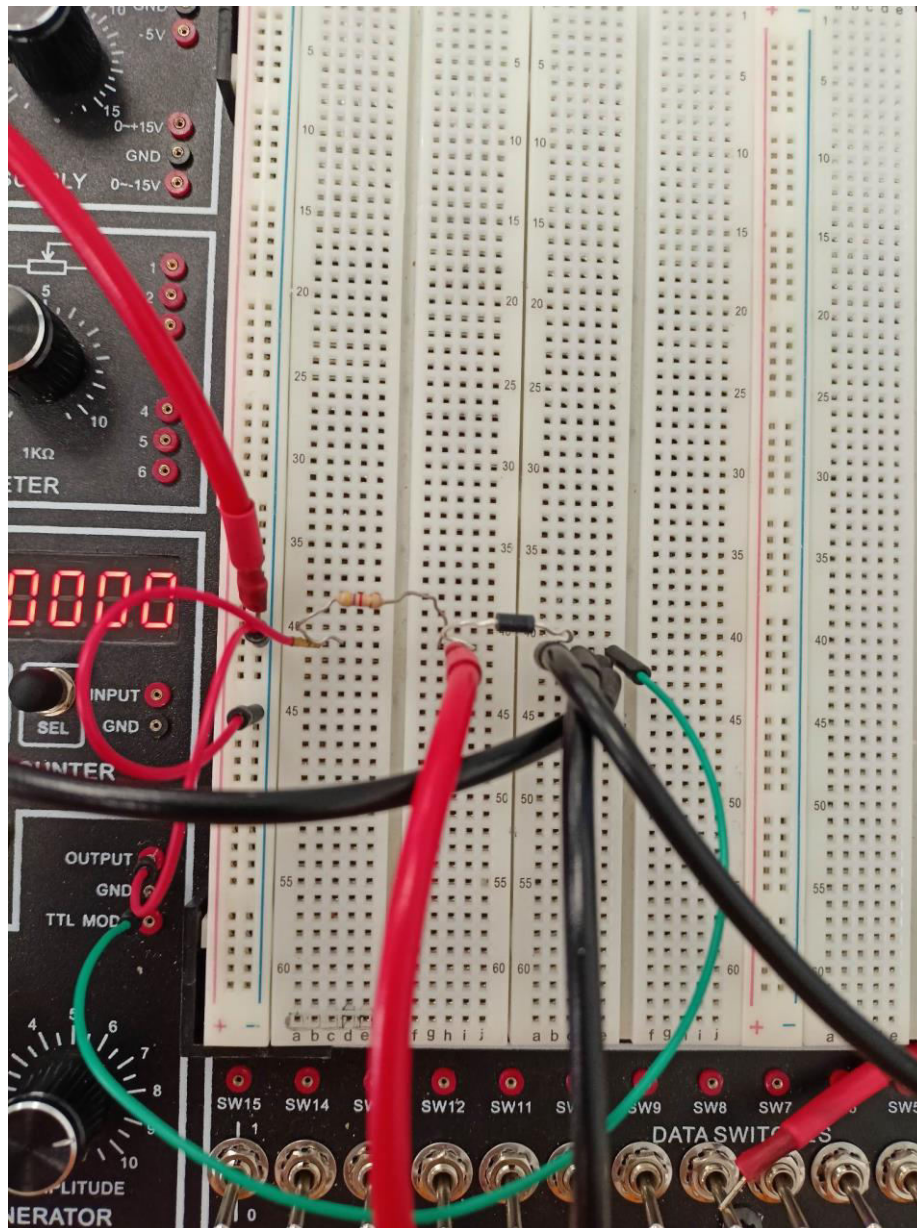
## Experimental Solution

Figure 3.8 shows the circuit of the simple rectification with diode as a switch and the anode pointing to ground, on the "breadboard", where it was carried out in the laboratory. The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 3.9).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 3.10).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 3.10).
4. We take channel A of the oscilloscope and connect the terminals to display the waveform on the screen. (Figure 3.10)

# ELECTRONICS

5. We set the switch to 2 Volts/Div. (Figure 3.11)
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 8 V<sub>p-p</sub> (Figure 3.11)
7. We align the waveform from the oscilloscope to be correctly positioned on the axes. (Figure 3.11)
8. We continue assembling the circuit by taking the 4.7 KOhm resistor and connecting it in series with the source (Figure 3.12).
9. We take the silicon diode and connect it in series with the resistor, with the anode connected to ground (Figure 3.13)
10. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the diode to obtain the output signal V<sub>max</sub> of the 1<sup>st</sup> column of Table 3 (Figure 3.14).
11. We also connect the ends of the digital multimeter bench set to calculate alternating and continuous voltage in parallel with the ends of the diode to obtain the voltage V<sub>μ</sub> of the 2<sup>nd</sup> column in Table 3 (Figure 3.15).



**Figure 3.8** Experimental illustration of simple rectification with a diode in the role of a switch, with the anode pointing towards ground on the "breadboard"

# ELECTRONICS



Figure 3.9 Step 1

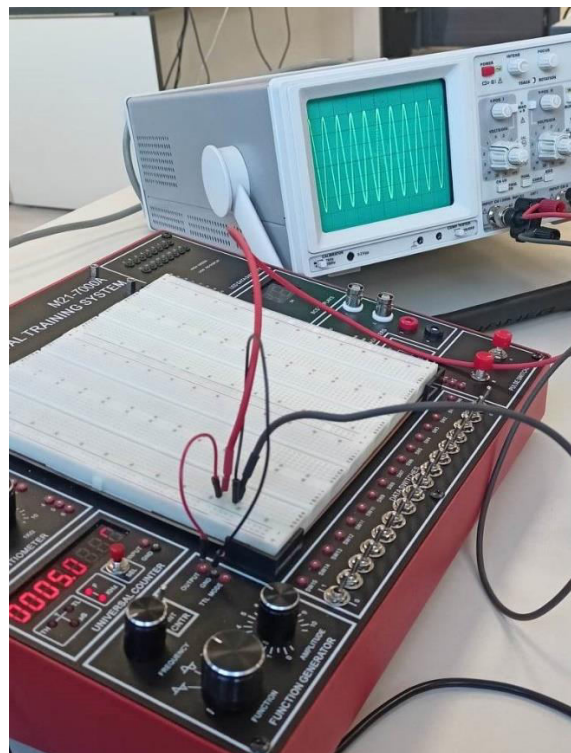


Figure 3.10 Step 2, 3, 4

# ELECTRONICS

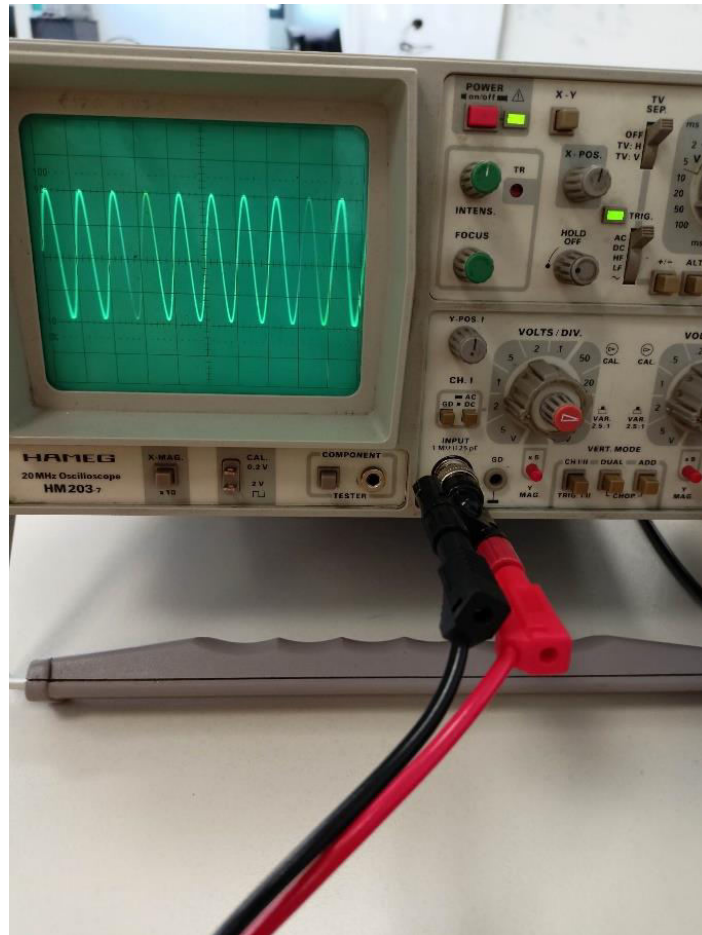


Figure 3.11 Step 5, 6, 7

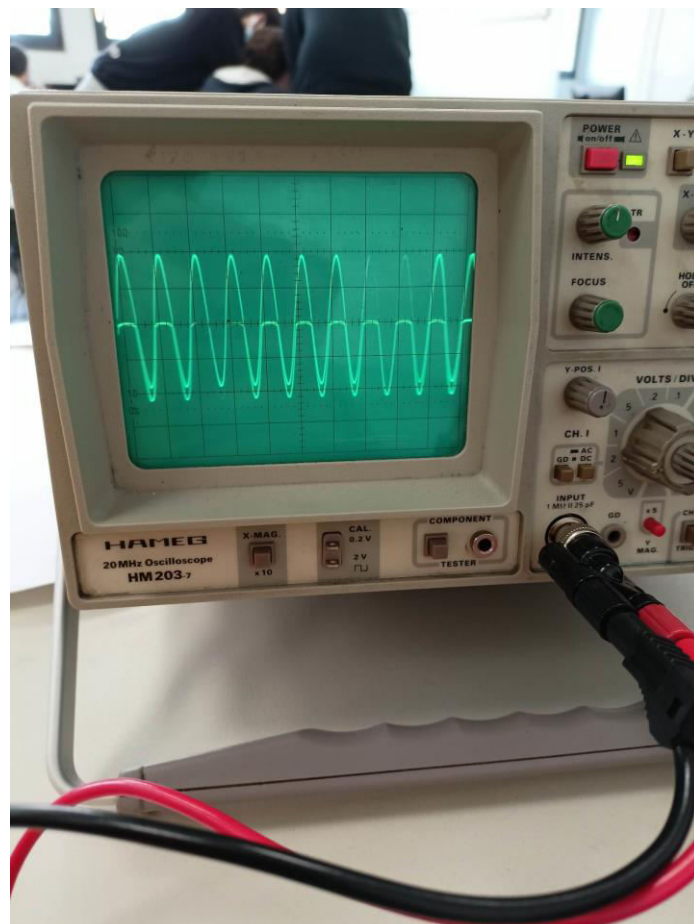


Figure 3.12 Step 8

# ELECTRONICS

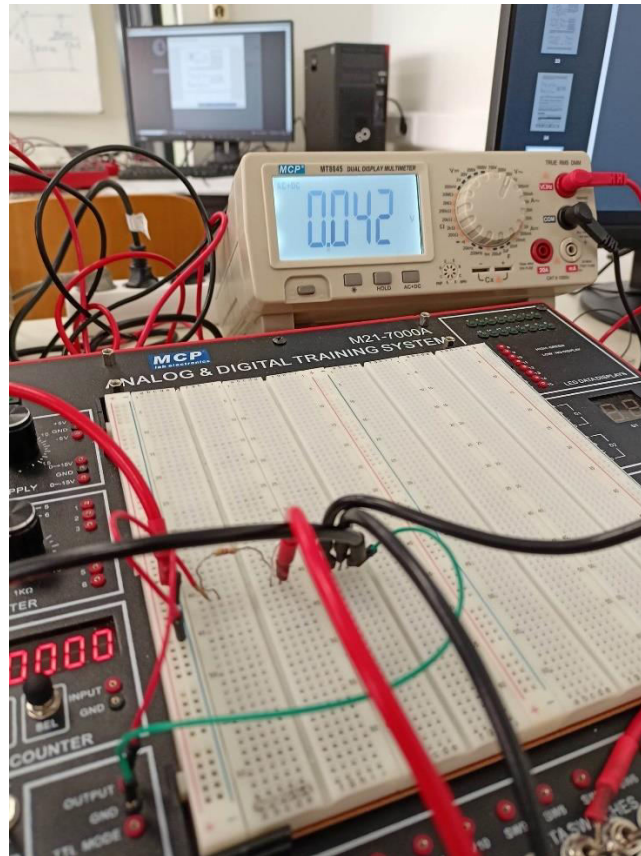


**Figure 2.13 Step 9**



**Figure 3.14 Step 10**

# ELECTRONICS



**Figure 3.15 Step 11**

In Figure 3.15, it is observed that the voltmeter reading is at 0.042 Volt and this is due to the fact that the diode is not connected correctly on the "breadboard" so the reading is wrong (verified in the "Simulation Solution" in Figure 3.6).

## **4. DOUBLE RECTIFICATION WITH BRIDGE**

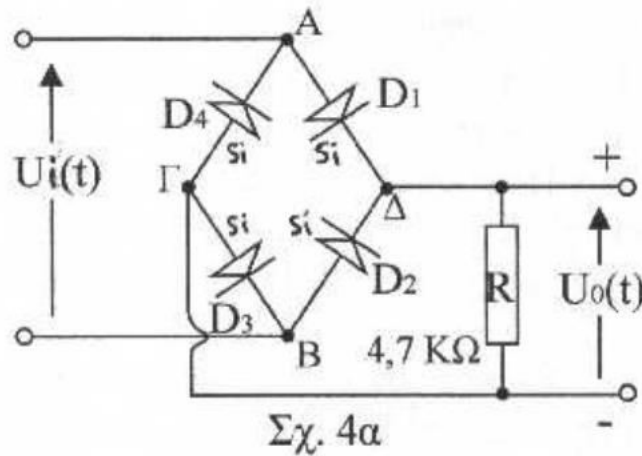
### **General**

The chapters "Simulation Solution" and "Experimental Solution" present in detail the components used, the recorded measurements and the observations made during the experiment with snapshots from the simulation software "Multisim" and photographs of the experiment from the laboratory environment respectively, while the chapter "Theoretical Solution" analyses the mathematical proof of the experiment.

### **Theoretical Solving**

Figure 4.1 shows the schematic illustration of the double lift bridge experiment.

# ELECTRONICS



**Figure 3.1** Schematic illustration of the double lift with bridge

Table 4 records the measurements in terms of the voltage  $V_o$  to  $V_p$  calculated by the oscilloscope between the ends of the resistor, the voltage  $V_\mu$  calculated by the voltmeter at these points, an average calculated value of  $2V_o / \pi$  and the frequency  $F_{out}$  of the output signal. It is worth noting that an alternating voltage of 8  $V_{p-p}$  (4 V)<sub>p</sub> and a frequency of 1 KHz was applied to the circuit.

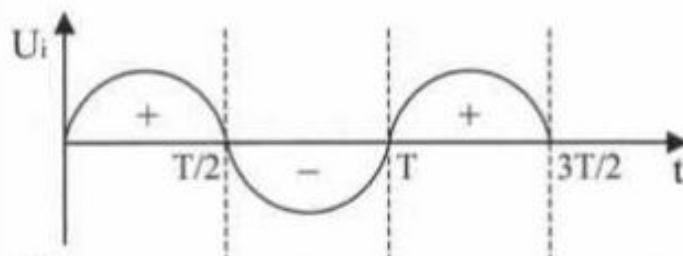
**Table 4**

| $V_\mu$ (Voltmeter in V) | $F_{out}$ (Hz) | $V_{max}$ (Oscilloscope in V) <sub>p</sub> | $V_\mu$ (V) = $2V_{max} / \pi$ |
|--------------------------|----------------|--------------------------------------------|--------------------------------|
| 1.024                    | 4              | 3                                          | 1.9                            |

The average value of the 4<sup>th</sup> column was calculated as follows :  $V_\mu = \frac{2 * V_{max}}{\pi} \rightarrow V_\mu = \frac{2 * 3}{3.14} \rightarrow V_\mu = 1.9$  Volt and is less than the voltage calculated by the voltmeter in the 2<sup>nd</sup> column (1.5 Volt).

The output frequency  $F_{out}$  was calculated from the period which is 0.5 sec (see Figure 4.) and from the formula  $F_{out} = \frac{1}{T} \rightarrow F_{out} = \frac{1}{0.5} \rightarrow F_{out} = 4$  Hz

For the input signal  $V_i$  this waveform is formed as shown in Figure 4.2. As observed, it is a sinusoidal voltage with a period T supplied by an AC voltage source.



**Figure 4.2** Waveform of the input signal  $V_i$

For the output signal  $V_o$  this waveform is formed as shown in Figure 4.3. As can be observed, during the positive half-period of the input signal  $V_i$ , point A is positive, while B is negative. Therefore, the following apply:

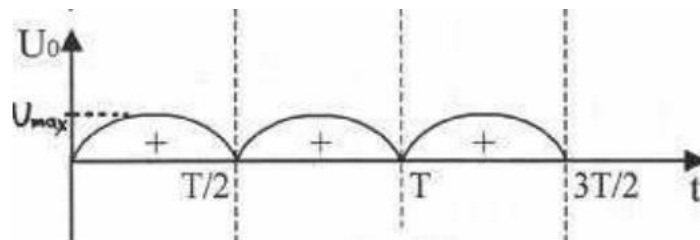
1. The anode of diode D1 is positively charged and the voltage is higher than that of the cathode, so D1 is turned off.
2. The cathode of diode D4 is positively charged and the voltage is higher than that of the anode, so diode D4 does not turn off.
3. The cathode of diode D3 is negatively charged and the voltage is less than that of the anode, therefore diode D3 is de-energized.
4. The anode of diode D2 is negatively charged and the voltage is lower than that of the cathode, so diode D2 does not turn on.

# ELECTRONICS

In contrast, during the negative half-period of the input signal  $V_i$ , point A is negative, while B is positive. Therefore, the following are valid:

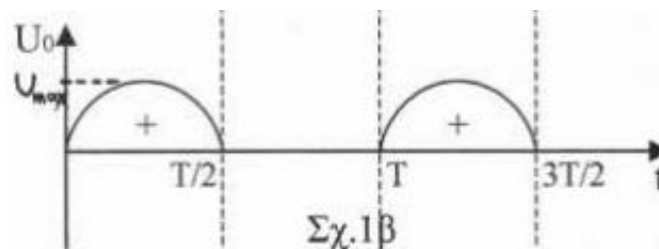
1. The anode of diode D1 is negatively charged and the voltage is less than that of the cathode, so D1 does not turn on.
2. The cathode of diode D4 is negatively charged and the voltage is less than that of the anode, therefore diode D4 is de-energized.
3. The cathode of diode D3 is positively charged and the voltage is higher than that of the anode, so diode D3 does not turn off.
4. The anode of diode D2 is positively charged and the voltage is higher than that of the cathode, so diode D2 is de-energized.

Therefore, with at least two silicon diodes orthogonally polarized in each half-period (D1, D3 in the positive and D2, D4 in the negative) the output voltage  $V_o$  is the same for each half-period.



**Figure 4.3** Waveform of the input signal  $V_o$

If diode D2 is disabled this means that the voltage will not be the same for each half period. In particular, at the positive half-period D1 and D3 remain correctly polarized, so the output signal  $V_o$  is a sinusoidal signal, while D4 is correctly polarized at the negative half-period as long as D1 and D3 are inversely polarized. Only D4 being grounded opposite inversely polarized D1 and D3 leads the circuit from the double rectification that existed, to single rectification (Figure 4.4).

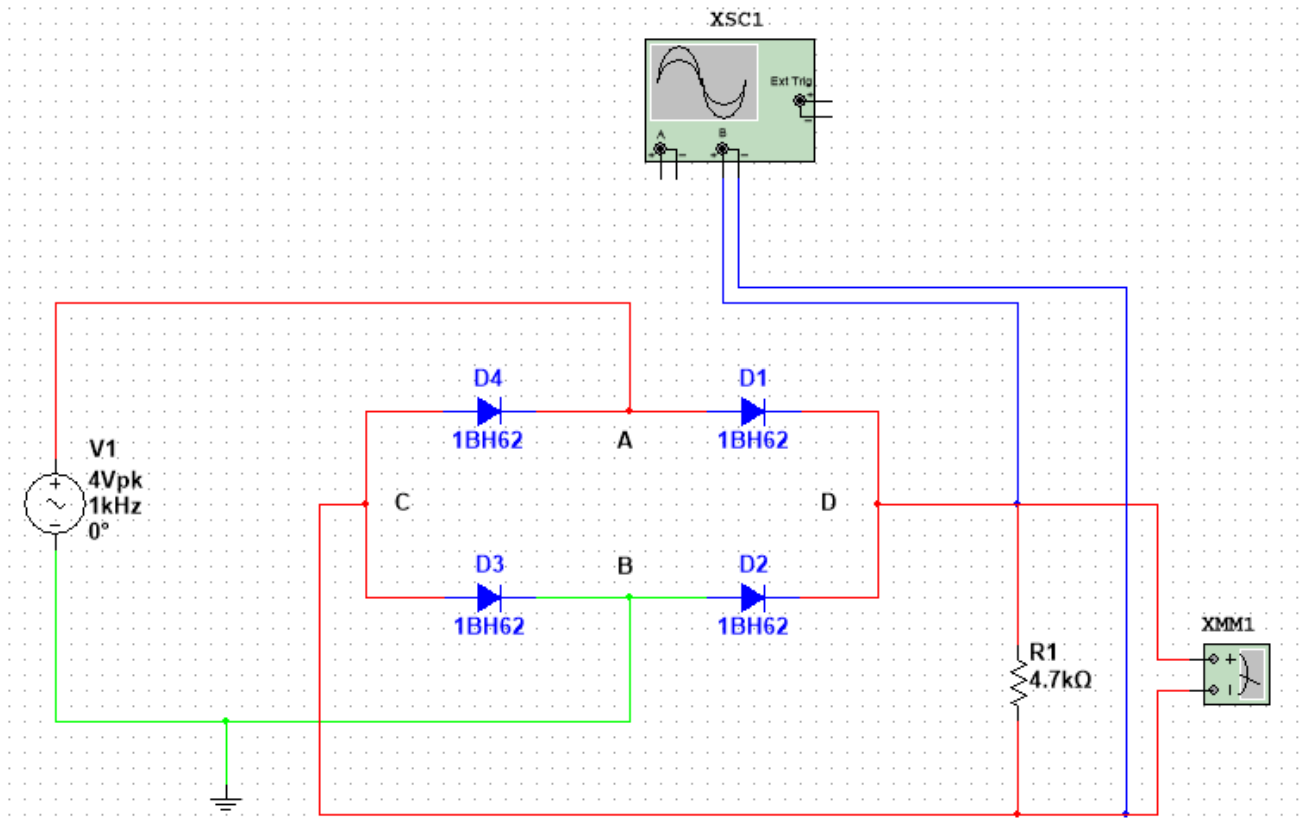


**Figure 4.4** Waveform of the input signal  $V_o$  with diode D2 turned off

## Simulative Resolution

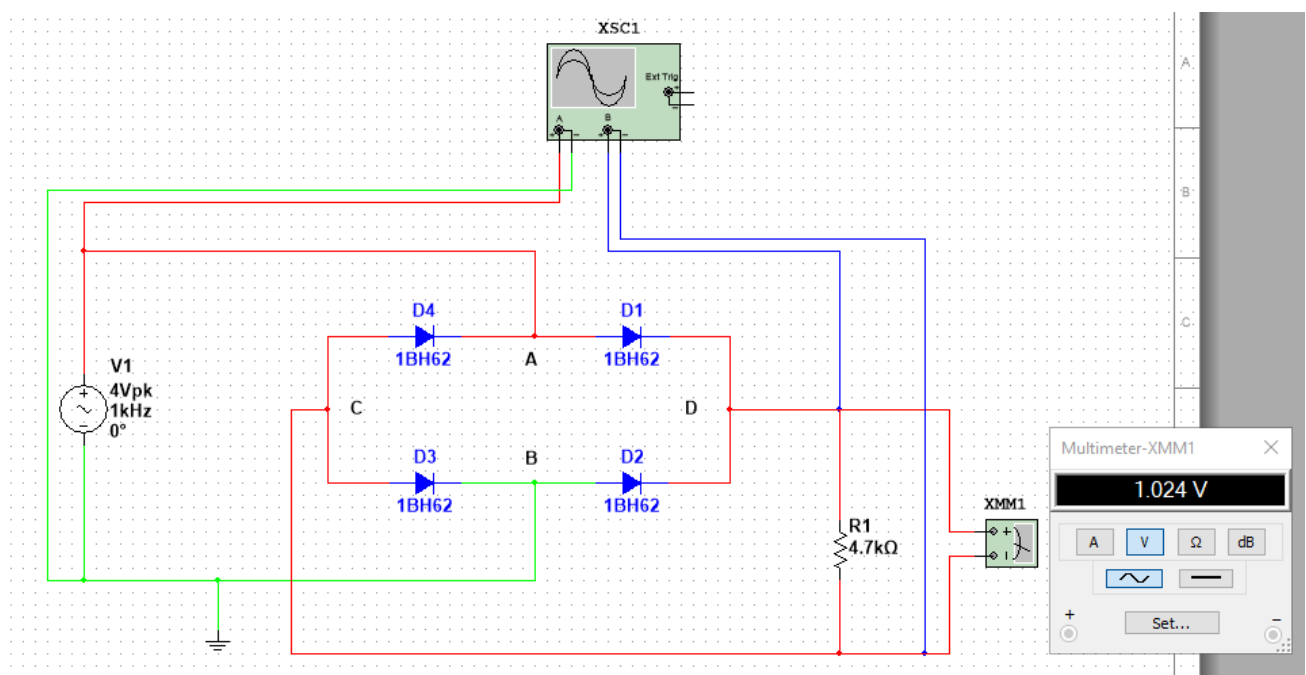
Figure 4.5 shows the simulated representation of the double rectification circuit with four diodes in a bridge shape in the simulation software "Multisim".

# ELECTRONICS



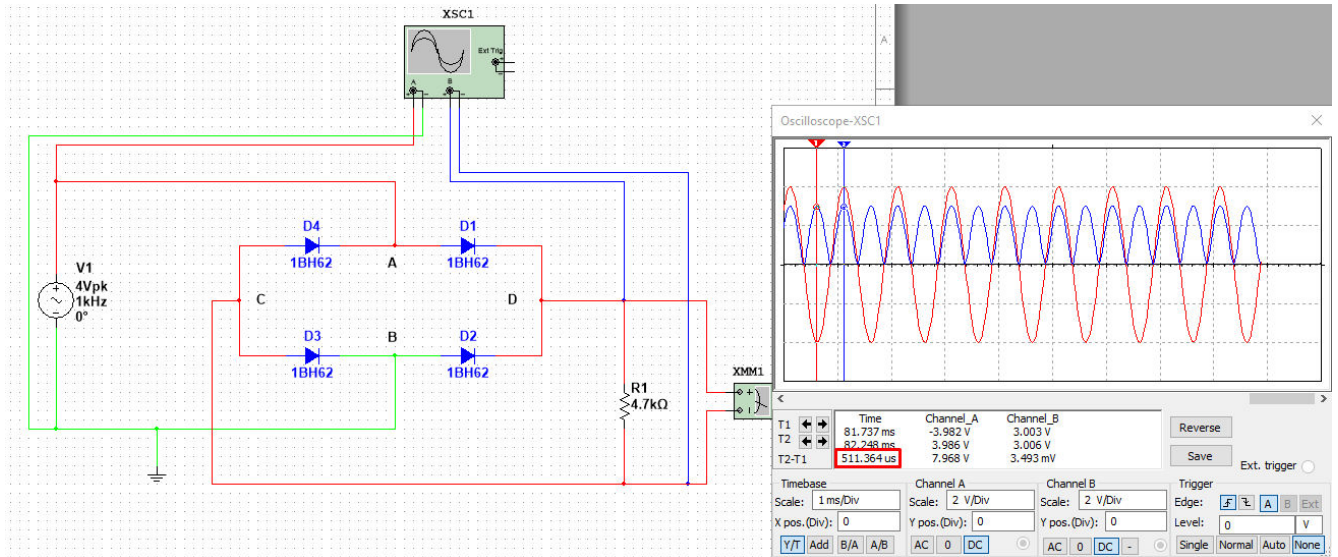
**Figure 4.5** Simulation of a double lift with four diodes in a bridge shape in "Multisim"

In Figures 4.6, 4.7, 4.8 and 4.9, the recorded measurements of Table 4 are simulated in the simulation software.

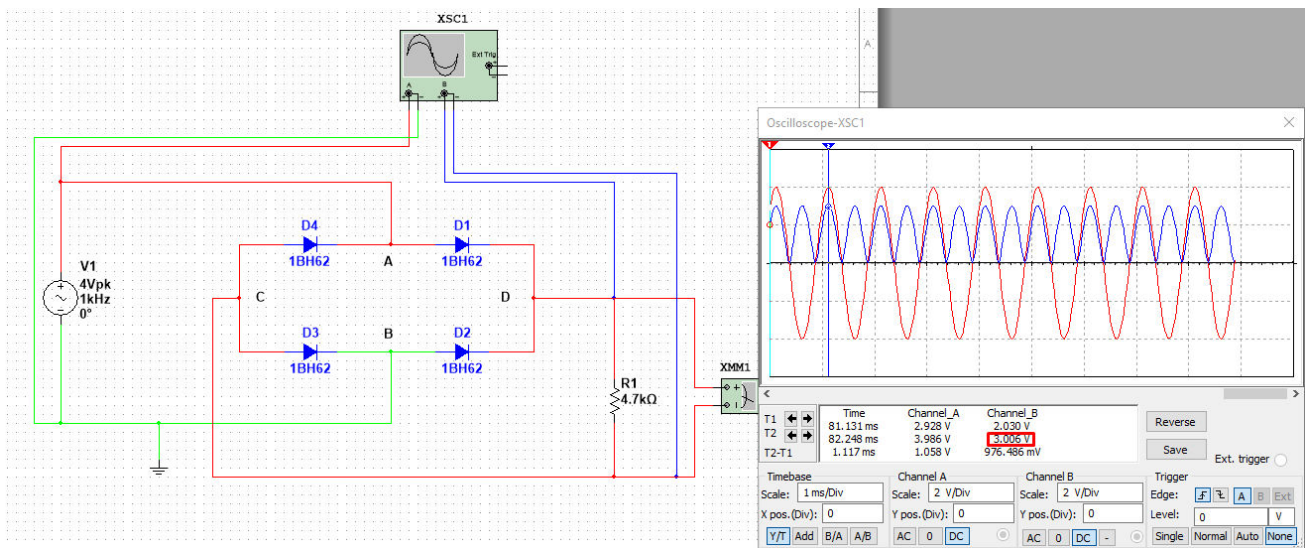


**Figure 4.6** Calculation of  $V_{\mu}$  of 1<sup>ns</sup> column with the voltmeter in "Multisim"

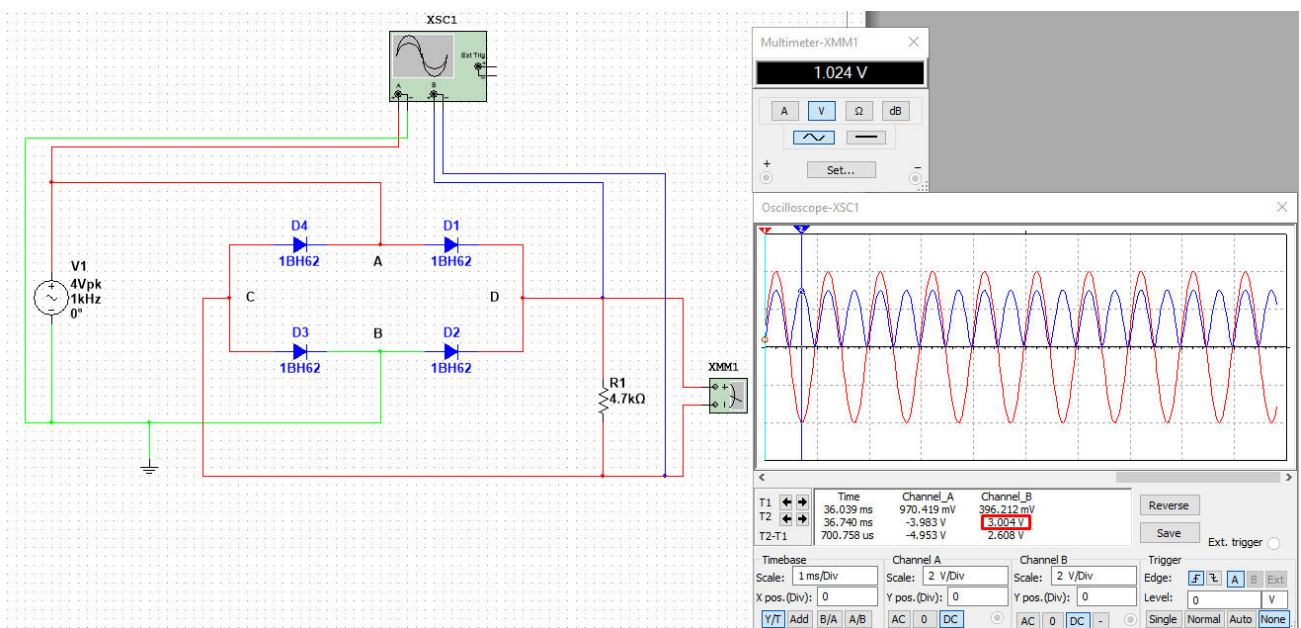
# ELECTRONICS



**Figure 4.7** Calculation of  $F_{out}$  of the 2<sup>nd</sup> column with the oscilloscope in "Multisim"

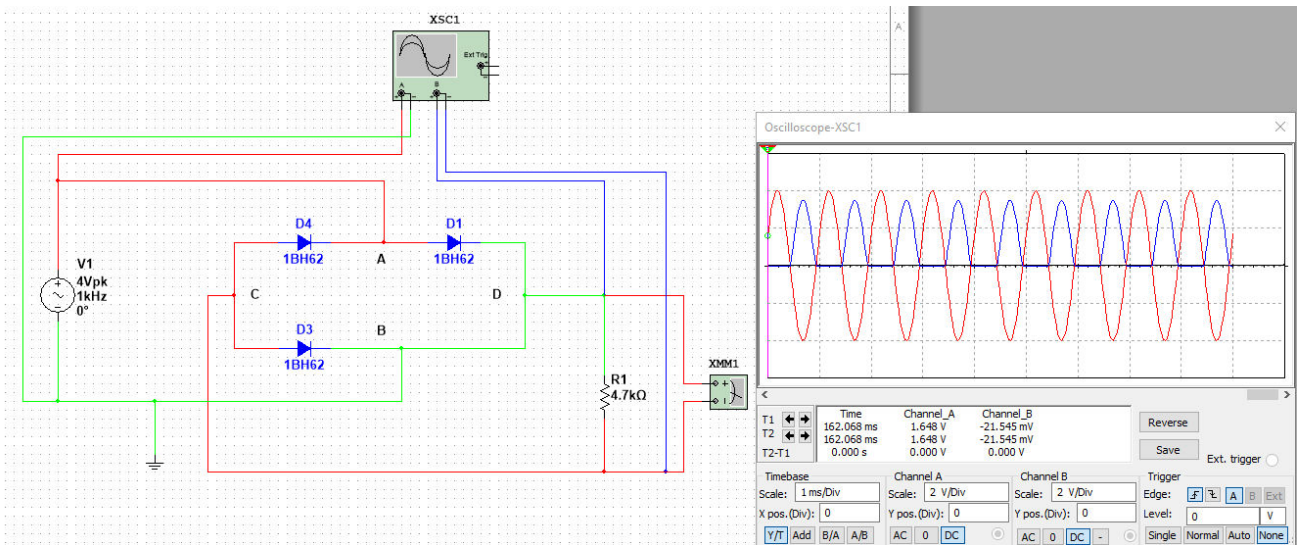


**Figure 4.8** Calculation of  $V_{max}$  of the 3<sup>rd</sup> column with the oscilloscope in "Multisim"



**Figure 4.9** Calculation of  $V_{\max}$  of the 4<sup>th</sup> column with the voltmeter and oscilloscope in "Multisim"

Figure 4.10 shows the simulation of the same circuit and the output signal  $V_o$  with diode D2 disabled, in the simulation software "Multisim".



**Figure 4.10** With diode D2 disabled the circuit is driven from double rectification to single rectification

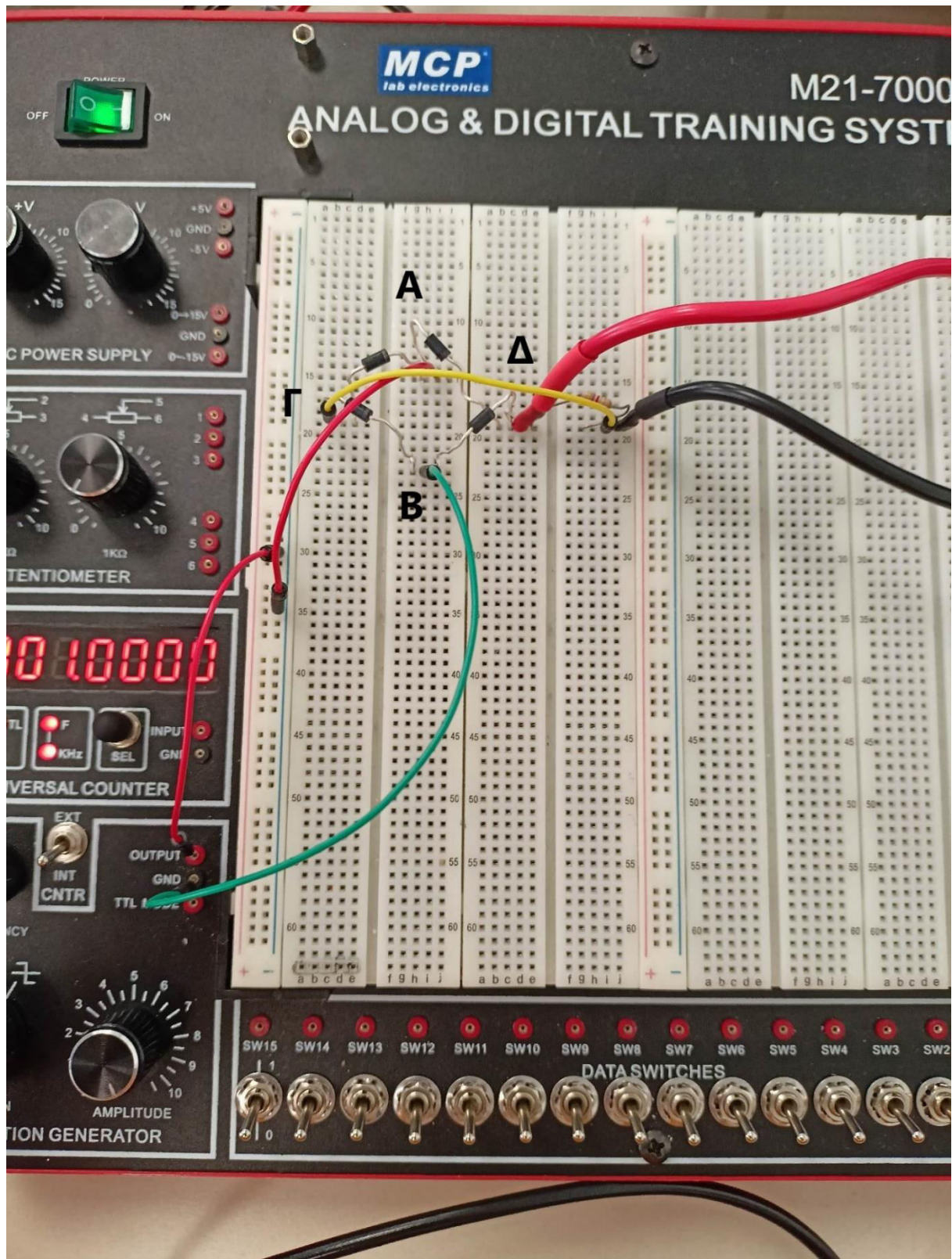
## Experimental Solution

Figure 4.11 shows the circuit of the double rectification with four diodes in a bridge shape, on the "breadboard", where it was carried out in the laboratory.

The construction steps of the experiment are as follows:

1. We set the input frequency to 1000 Hz (Figure 4.12).
2. We take a cable and connect the "Output" of the "breadboard", which is the alternating signal, to a power supply hole (Figure 4.13).
3. Together with the power supply we connect the grounding cable to the "breadboard" (Figure 4.13).
4. We take channel A of the oscilloscope and connect the terminals to display the waveform on the screen. (Figure 4.13).
5. We set the switch to 2 Volts/Div. (Figure 4.14.)
6. We adjust the amplitude of the waveform from the "breadboard" so that the input signal is 8  $V_{p-p}$  (Figure 4.14).
7. We align the oscilloscope so that the waveform is correctly positioned on the axes (Figure 4.14).
8. We take the four silicon diodes and connect them on the "breadboard" in a bridge shape (Figure 4.15).
9. Connect the source to the bridge at point A between diodes D1 and D4 with a cable (Figure 4.16).
10. Connect the grounding wire to the bridge at point B between diodes D2 and D3 (Figure 4.17).
11. Connect the resistor in series with the bridge and one end at point D between diodes D1 and D2 (Figure 4.18).
12. With a cable we connect the other end of the resistor to point C between diodes D3 and D4 (Figure 4.19).
13. We also take channel B of the oscilloscope and connect it in parallel with the terminals of the resistor to obtain the output signal  $V_{\max}$  of the 3<sup>rd</sup> column of Table 4 (Figure 4.20).
14. We also connect the ends of the digital multimeter bench set to calculate alternating and continuous voltage in parallel with the ends of the resistor to obtain the voltage  $V_{\mu}$  of the 1<sup>st</sup> column in Table 4 (Figure 4.21).

# ELECTRONICS



**Figure 4.11** Experimental illustration of double rectification with four bridge-shaped silicon diodes on the breadboard

# ELECTRONICS



Figure 4.12 Step 1

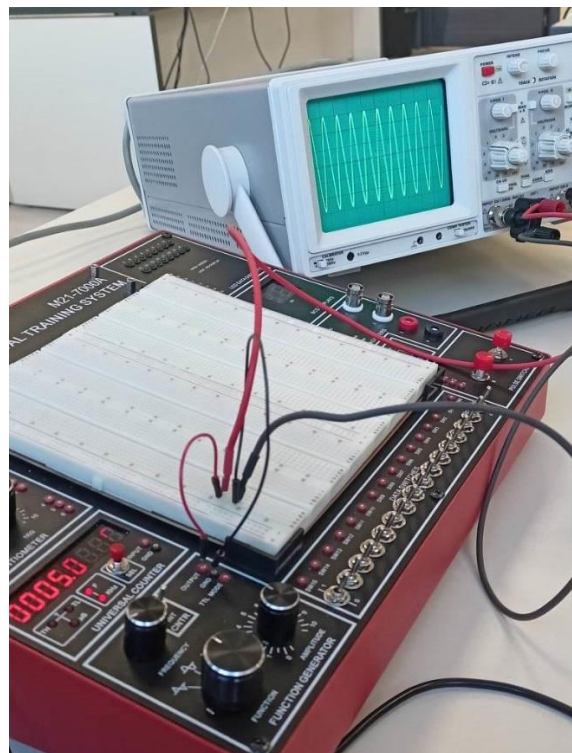


Figure 4.13 Step 2, 3, 4

# ELECTRONICS

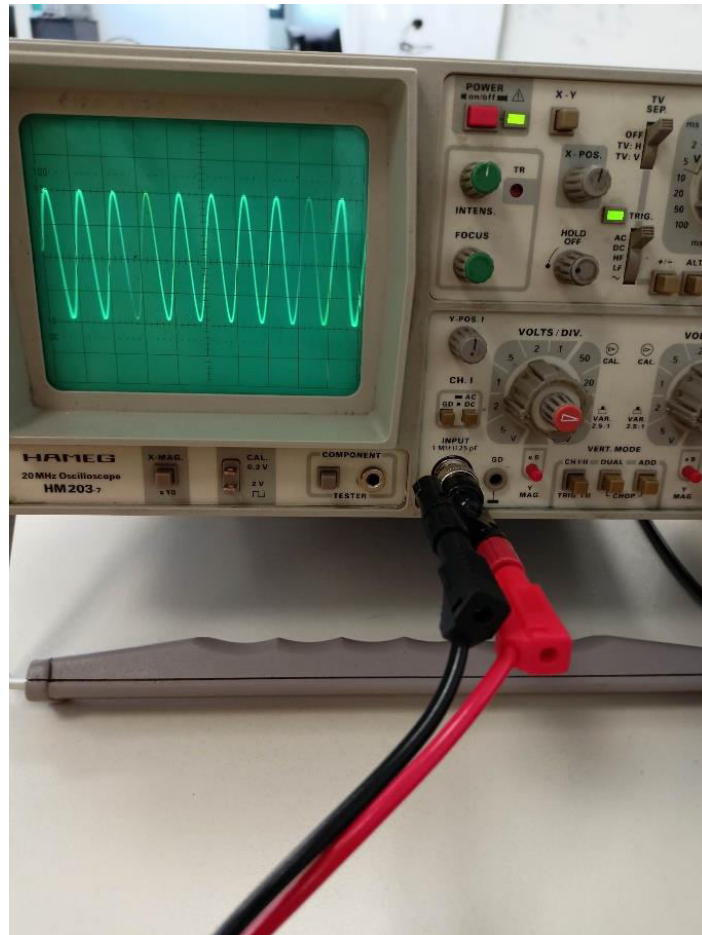


Figure 4.14 Step 5, 6, 7

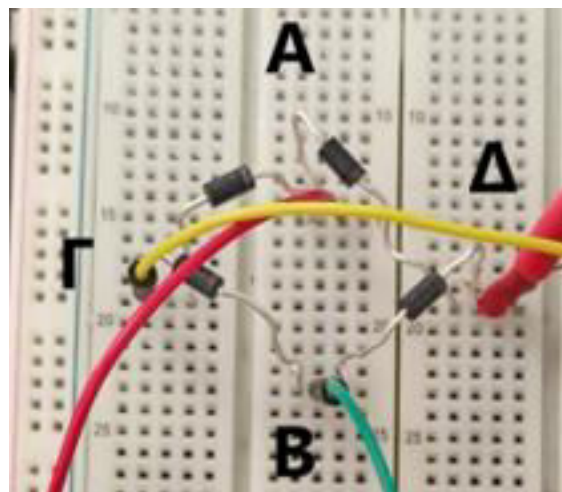


Figure 4.15 Step 8

# ELECTRONICS

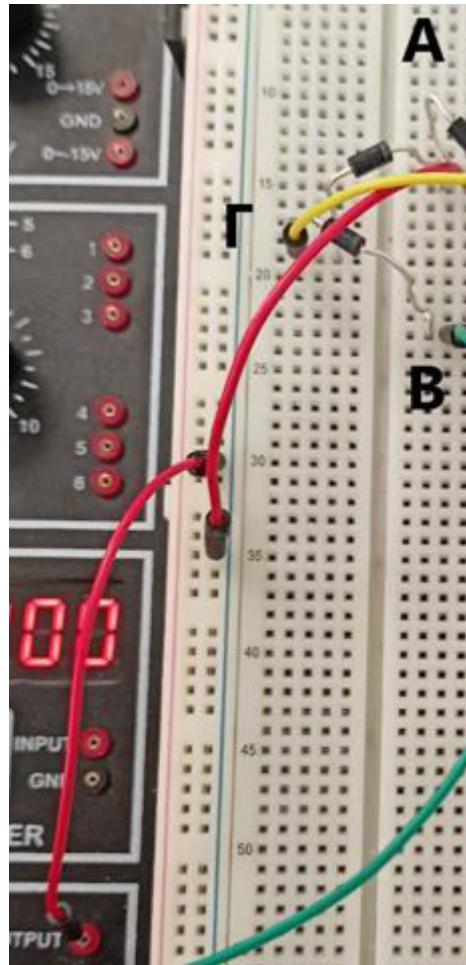


Figure 4.16 Step 9

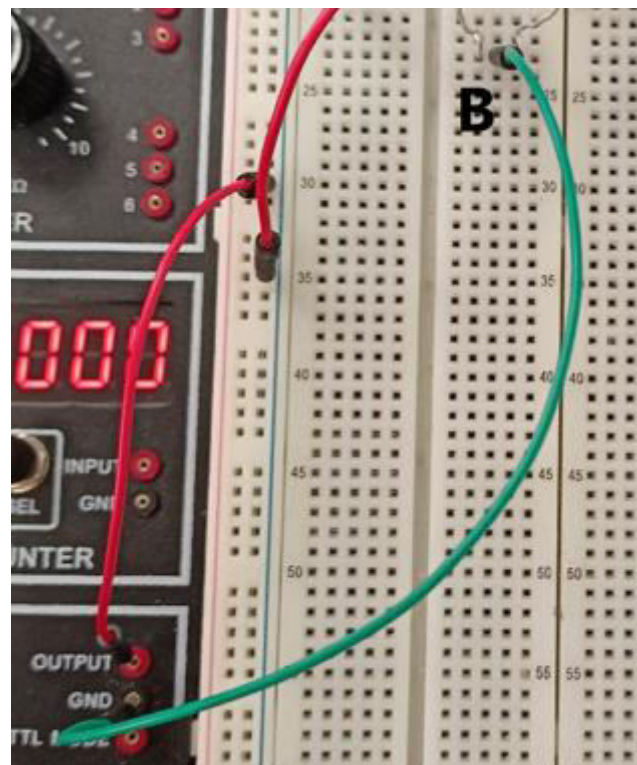


Figure 4.17 Step 10

# ELECTRONICS

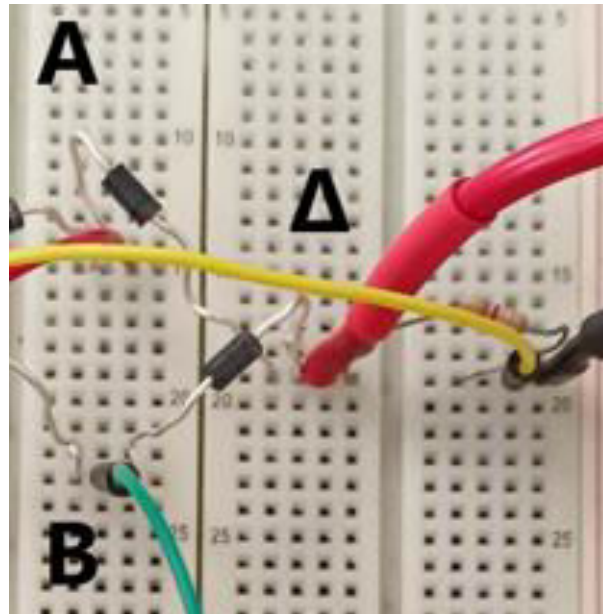


Figure 4.18 Step 11

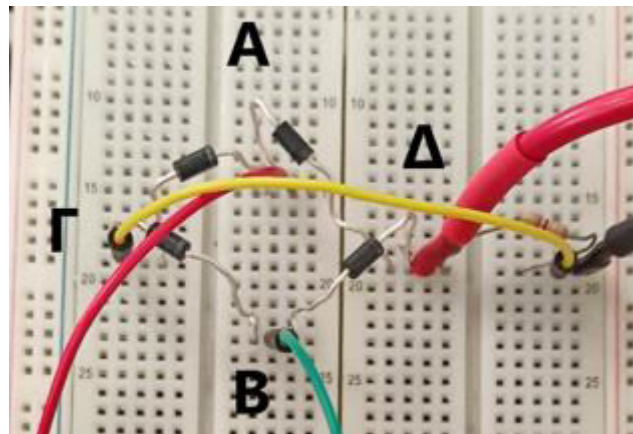


Figure 4.19 Step 12

# ELECTRONICS

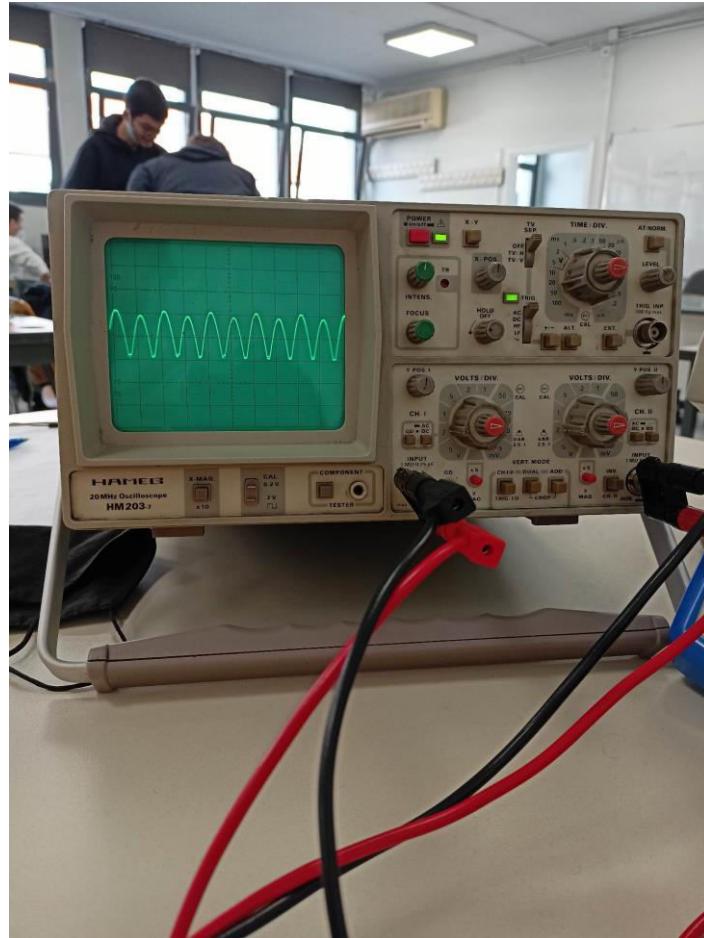


Figure 4.20 Step 13

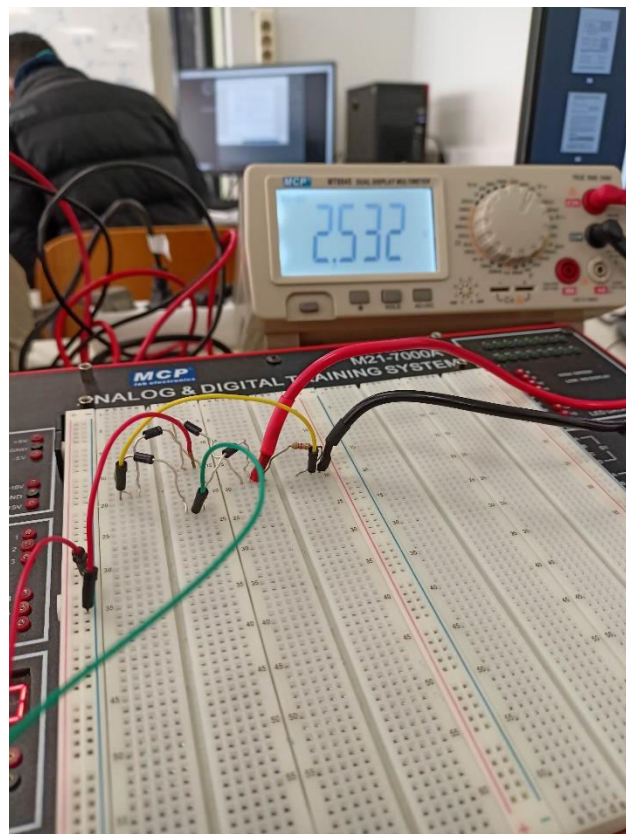


Figure 4.21 Step 14

# ELECTRONICS

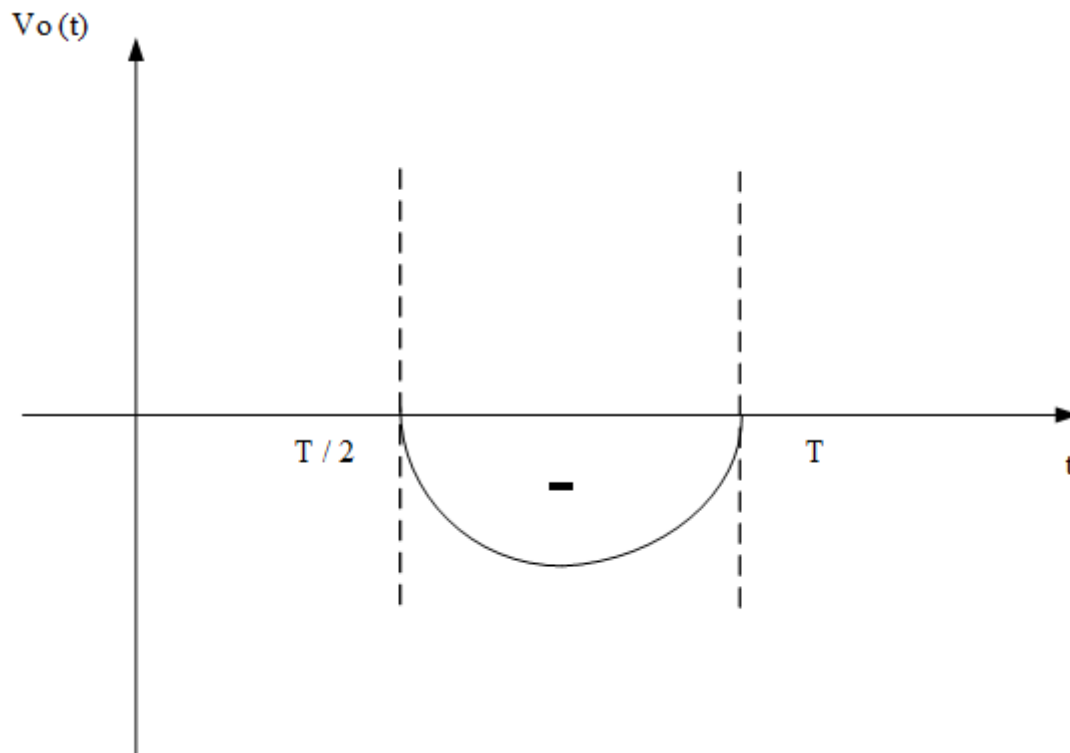
In Figure 4.21. it is observed that the voltmeter reading is at 2.532 Volts, while the reading in the simulated solution is 1.042 Volts (Figure 4.6). This is due to the fact that the voltmeter is not connected to the "breadboard" at the signals requested to calculate the voltage between the ends of the resistor, and as a result the reading is not the requested reading.

## 5. QUESTIONS

### Question 5.2

*If we placed the diode in the circuit in Figure 1.1 upside down, what would the output waveform be and why?*

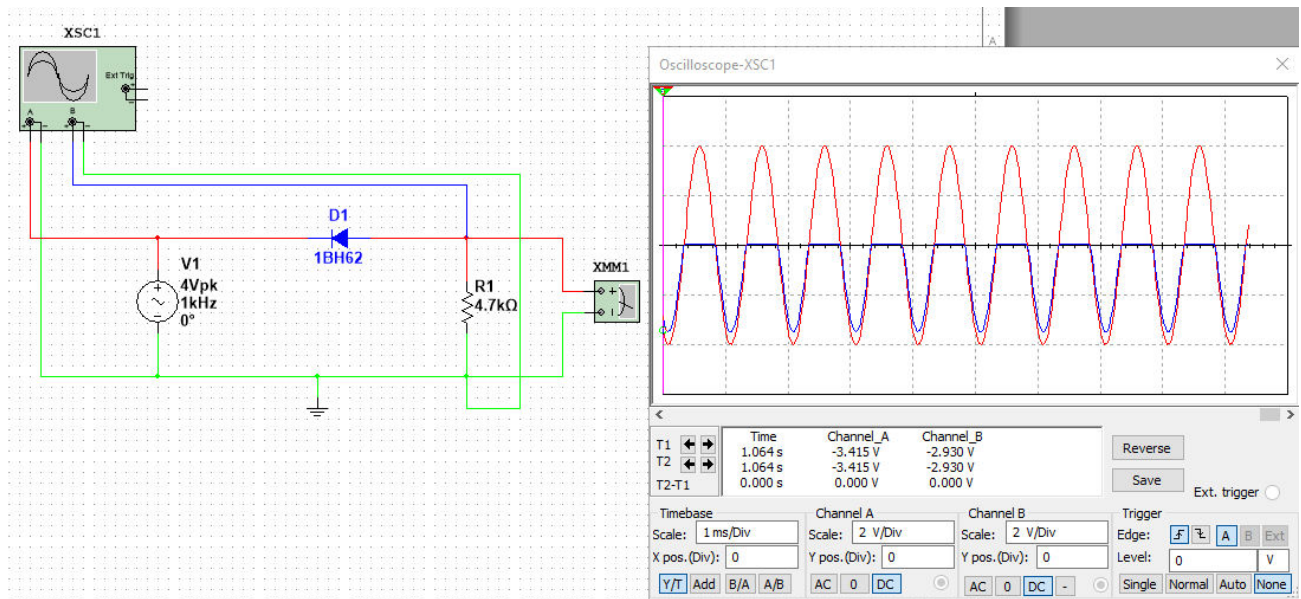
By reversing the polarity of the diode of the circuit of Figure 1.1, it would have the consequence that the diode would turn off this time, during the negative half-period of the input signal  $V_i$ , as, during the positive half-period, the cathode is positively charged, resulting in a higher voltage than that of the anode. Thus, the diode would be inversely polarized during the positive half period and normally polarized during the negative half period. The output waveform would be as follows :



**Figure 5.1** Output waveform of the single resistor rectifier with the diode pointing towards the source

Figure 5.2 shows the simulated solution of the circuit in the simulation software "Multisim"

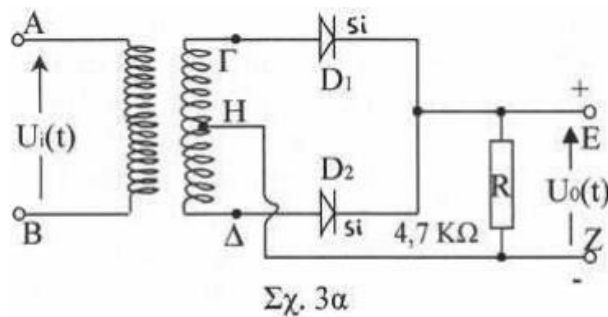
# ELECTRONICS



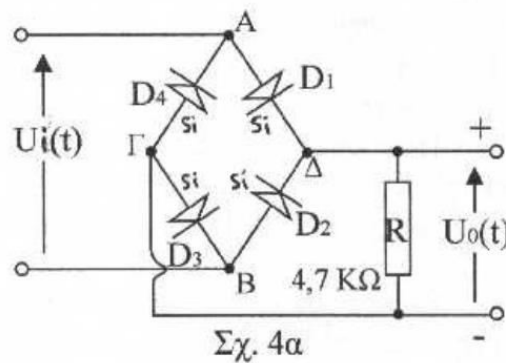
**Figure 5.2** Output waveform of the simple rectification with resistor with the diode facing the source, in "Multisim"

## Question 5.3

What is the difference between circuits 3a and 4a and why?



**Figure 5.3** Circuit 3a



**Figure 5.4** Circuit 4a

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Circuits 3a and 4a, although both achieve full or double rectification, have differences.

1. The circuit of Figure 5.3 achieves voltage switching on signals C and D through a coil winding with the ultimate goal that in one half-period one diode (D1) connected to point C is turned on and in the other half-period the second diode (D2) connected to point D is turned on. Thus, the signal is rectified in each half-period alternately by the two diodes. The same is achieved in the bridge circuit of Figure 5.4 with the difference that here there are pairs of diodes that are simultaneously orthogonally polarized at one half-period. Also, no coil winding is interposed.
2. In the circuit of Figure 5.4, the output signal  $V_o$  shows a larger voltage drop than the circuit of Figure 5.3 with one right-polarized diode in one half-period, due to two right-polarized diodes in one half-period.

## **Question 5.4**

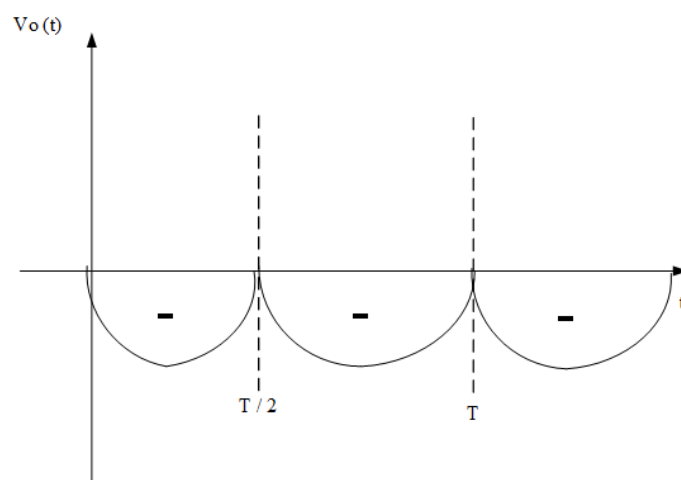
*Where are rectification circuits used and why?*

Rectifying circuits are used in electronic devices that use transistors or integrated circuits because of the need for consistent voltages resulting from rectification.

## **Question 5.6**

*If we placed the diodes of the circuit in Figure 5.3 upside down then what would the output waveform be and why? Draw it.*

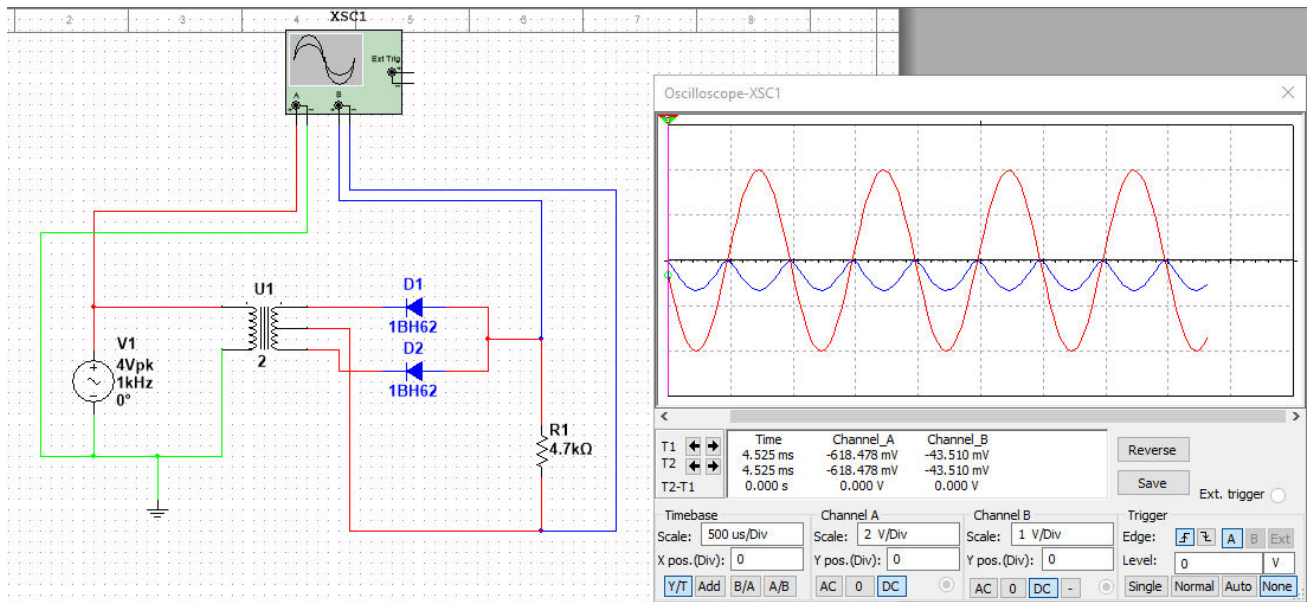
By reversing the polarity of both diodes (D1, D2) of the circuit in Figure 5.3, it would result in the diodes rectifying for negative voltages. This is due to the fact that the diode rectifying the negative alternations of the input signal  $V_i$ , would be the first to be correctly polarized, resulting in the next alternation being followed by the diode rectifying the positive alternations. The output waveform would be as follows :



**Figure 5.5** Output waveform of the double rectification with a mid-range transformer with the diodes reversed

Figure 5.6 shows the simulated solution of the circuit in the simulation software "Multisim"

# ELECTRONICS



**Figure 5.6** Output waveform of the double rectification with a mid-range transformer with the diodes reversed, in "Multisim"

# ELECTRONICS



Thank you for your attention.

